

LAB Project Nº 1 - VLAN/STP/OSPF MONO-AREA

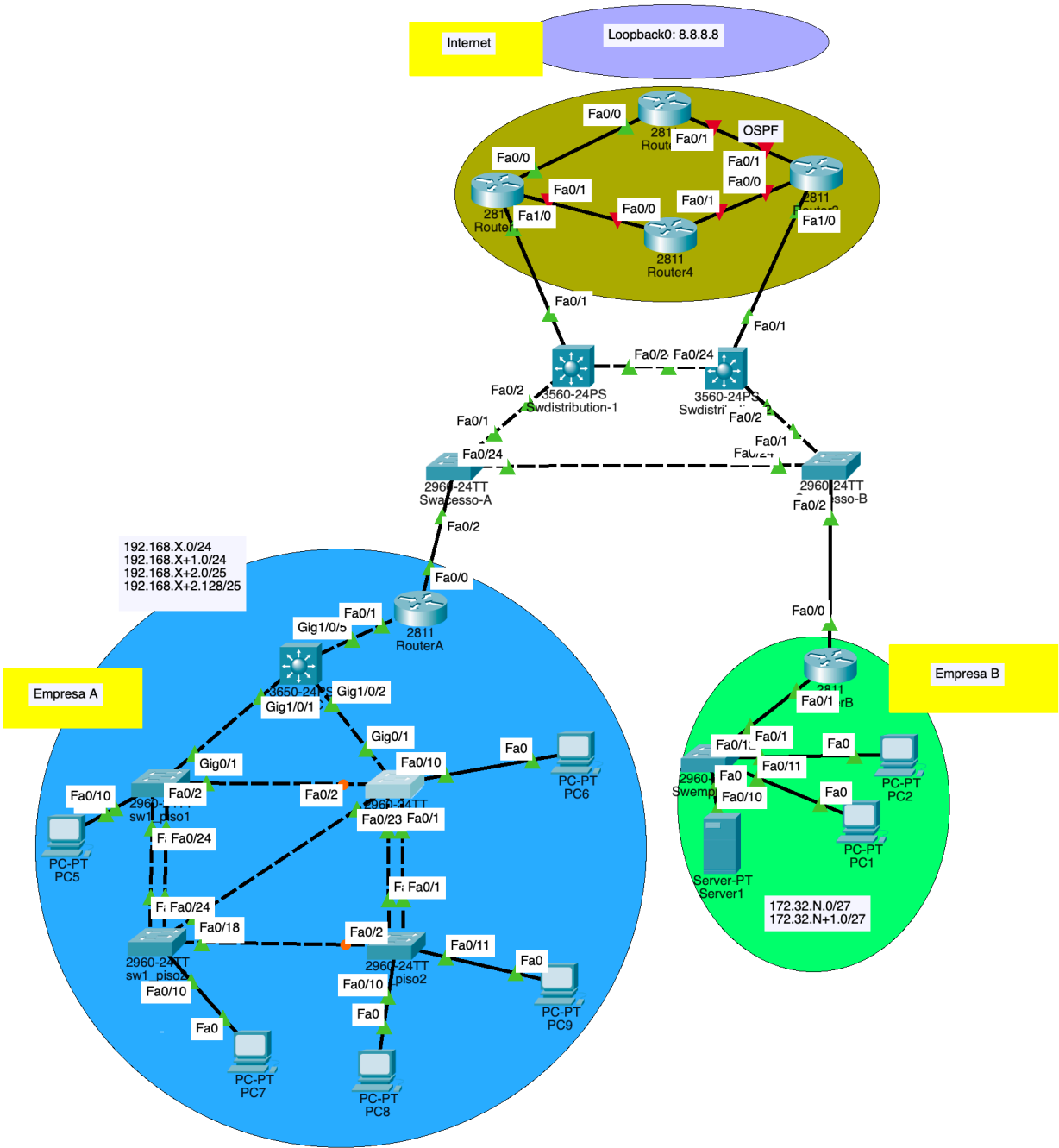
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Group 1 - Class 51D

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Introduction

This networking project successfully demonstrates the end-to-end implementation of enterprise-grade network infrastructure, progressing systematically from basic Layer 2 segmentation to advanced dynamic routing with public addressing. The project validates theoretical concepts through practical implementation across multiple network domains, establishing a robust, scalable, and secure infrastructure meeting all specified business requirements.

It served as a comprehensive practical demonstration of networking fundamentals:

Protocol Mastery

- **STP/RSTP:** Loop prevention and path optimization in Layer 2 networks
- **OSPF:** Dynamic routing, neighbor relationships, and path calculation
- **VLAN Trunking:** 802.1Q encapsulation and trunk management
- **Subnetting:** Efficient address space utilization and hierarchical design

Troubleshooting Proficiency

- Methodical fault isolation across OSI layers
- Effective use of show and debug commands
- Systematic validation of connectivity and policies
- Documentation of failure scenarios and resolutions

Design Principles

- Hierarchical network design
- Redundancy and high availability
- Security through segmentation
- Scalability and future-proofing

To accomplish this we will configure and simulate a network between two enterprises (A and B) and an ISP (Internet Service Provider).

The Enterprise A will be constituted of 1 router, 5 switches, 5 PCs and 3 VLANs, VLAN 11 (Accounting), VLAN 12 (Secretariat) and VLAN 13 (Computer_science).

The Enterprise B will be constituted of 1 router, 1 switch, 1 server, 1 pc and 2 VLANs, VLAN 2 (Servers) and VLAN 3 (Engineering).

The ISP will be constituted by 4 routers and 4 switches, that connects to "Internet" using a Loopback

1. Enterprise A - L2 Foundations (VLAN/STP/RSTP)

1.1 Implementation

1.1.1 VLANs

To configure the static VLANs according to the table in the point 4.2.1 it was necessary to remove in each switch the previous VLANs:

```
Switch(config)# no vlan 20
Switch(config)# no vlan 30
Switch(config)# no vlan 40
Switch(config)# no vlan 45
```

For security reasons, we assigned the `Native VLAN` to **VLAN 99** in all the switches:

```
Switch(config) vlan 99
Switch(config) interfaces range [desired interfaces]
Switch(config-if)# switchport trunk native vlan 99
Switch(config-if)# switchport trunk allowed vlan add 99
Switch(config-if)# switchport nonegotiate
```

To create and name our VLANs we used:

```
Switch(config)# vlan 11
Switch(config-vlan)# name Accounting
Switch(config)# vlan 12
Switch(config-vlan)# name Secretariat
Switch(config)# vlan 13
Switch(config-vlan)# name Computer_science
```

Then, interface configuration by switch:

```
# Switch SW_DC -----
# Gig1/0/1 -> sw1_piso1 - trunk
# Gig1/0/2 -> sw2_piso1 - trunk
# Gig1/0/5 -> Router A - trunk

SW_DC(config)# interface range gig1/0/1-2,gig1/0/5
SW_DC(config-if-range)# switchport mode trunk
SW_DC(config-if-range)# switchport trunk native vlan 99
SW_DC(config-if-range)# switchport trunk allowed vlan 11-13
SW_DC(config-if-range)# switchport nonegotiate

# Switch sw1_piso1 -----
# Fa0/2 -> sw2_piso1 - trunk
# Fa0/10 -> PC5 - access
# Fa0/23-24 -> sw1_piso2 - trunk
# Gig0/1 -> SW_DC - trunk

sw1_piso1(config)# interface range fa0/2,fa0/23-24,Gig0/1
sw1_piso1(config-if-range)# switchport mode trunk
sw1_piso1(config-if-range)# switchport trunk native vlan 99
sw1_piso1(config-if-range)# switchport trunk allowed 11-13
sw1_piso1(config-if-range)# switchport nonegotiate
sw1_piso1(config-if-range)#
sw1_piso1(config-if-range)# interface fa0/10
sw1_piso1(config-if)# switchport mode access
sw1_piso1(config-if)# switchport access vlan 12
sw1_piso1(config-if)# switchport nonegotiate

# Switch sw2_piso1 -----
# Fa0/1 -> sw2_piso2 - trunk
# Fa0/2 -> sw1_piso1 - trunk
# Fa0/10 -> PC6 - access
# Fa0/23 -> sw2_piso2 - trunk
# Fa0/24 -> sw1_piso2 - trunk
# Gig0/1 -> SW_DC - trunk

sw2_piso1(config)# interface range fa0/1-2,fa0/23-24,Gig0/1
sw2_piso1(config-if-range)# switchport mode trunk
sw2_piso1(config-if-range)# switchport trunk native vlan 99
sw2_piso1(config-if-range)# switchport trunk allowed 11-13
sw2_piso1(config-if-range)# switchport nonegotiate
sw2_piso1(config-if-range)#
sw2_piso1(config-if-range)# interface fa0/10
sw2_piso1(config-if)# switchport mode access
sw2_piso1(config-if)# switchport access vlan 13
sw2_piso1(config-if)# switchport nonegotiate

# Switch sw1_piso2 -----
# Fa0/2 -> sw2_piso1 - trunk
# Fa0/10 -> PC7 - access
```

```
# Fa0/18 -> sw2_piso2 - trunk
# Fa0/23-24 -> sw1_piso1 - trunk

sw1_piso2(config)# interface range fa0/2,fa0/18,fa0/23-24
sw1_piso2(config-if-range)# switchport mode trunk
sw1_piso2(config-if-range)# switchport trunk native vlan 99
sw1_piso2(config-if-range)# switchport trunk allowed 11-13
sw1_piso2(config-if-range)# switchport nonegotiate
sw1_piso2(config-if-range)#
sw1_piso2(config-if-range)# interface fa0/10
sw1_piso2(config-if)# switchport mode access
sw1_piso2(config-if)# switchport access vlan 11
sw1_piso2(config-if)# switchport nonegotiate

# Switch sw2_piso2 -----
# Fa0/1 -> sw2_piso1 - trunk
# Fa0/2 -> sw1_piso2 - trunk
# Fa0/10 -> PC8 - access
# Fa0/11 -> PC9 - access
# Fa0/24 -> sw2_piso1 - trunk

sw2_piso2(config)# interface range fa0/1-2,fa0/24
sw2_piso2(config-if-range)# switchport mode trunk
sw2_piso2(config-if-range)# switchport trunk native vlan 99
sw2_piso2(config-if-range)# switchport trunk allowed 11-13
sw2_piso2(config-if-range)# switchport nonegotiate
sw2_piso2(config-if-range)#
sw2_piso2(config-if-range)# interface fa0/10
sw2_piso2(config-if)# switchport mode access
sw2_piso2(config-if)# switchport access vlan 12
sw2_piso2(config-if)# switchport nonegotiate
sw2_piso2(config-if)#
sw2_piso2(config-if)# interface fa0/11
sw2_piso2(config-if)# switchport mode access
sw2_piso2(config-if)# switchport access vlan 11
sw2_piso2(config-if)# switchport nonegotiate
```

The only PC that uses **VLAN 13 - Computer_science** is the PC6, witch is connected to sw2_piso1 and this switch to SW_DC, but because of connection redundancy reasons, it was decided to add this VLAN to all the switches.

VLAN	NAME	IP GATEWAY	NETWORK	PCs
11	Accounting	172.20.11.254	172.20.11.0/24	PC7, PC9
12	Secretariat	172.20.12.254	172.20.12.0/24	PC5, PC8
13	Computer science	172.20.13.126	172.20.13.0/25	PC6

table 1 - Enterprise A VLANs

1.1.2 STP (RB/roles/costs)

Switches / Bridges Priorities (All VLANs)

Switch	Priority	MAC Address	Root MAC Address
SW_DC	32779	0002.4AE4.E093	same
sw1_piso1	32779	0009.7C65.BA4B	0002.4AE4.E093
sw2_piso2	32779	00E0.8FE1.53D7	0002.4AE4.E093
sw1_piso2	32779	00E0.A3CE.4A46	0002.4AE4.E093
sw2_piso1	32779	00E0.F950.631A	0002.4AE4.E093

table 2 - Switches / Bridges Priorities (All VLANs)

VLAN 11 (Accounting) - Root Bridge, Port Roles and Root Port Costs

Switch	Port	PC	RPC	RP	DP	BLK	Notes
SW_DC	Gig1/0/1	4	-				
SW_DC	Gig1/0/2	4	-				
SW_DC	Gig1/0/5	19	-				to Fa port at RouterA
sw1_piso1	Gig0/1	4	4	X			
sw1_piso1	Fa0/2	19	19		X		
sw1_piso1	Fa0/23	19	19		X		
sw1_piso1	Fa0/24	19	19		X		
sw2_piso2	Fa0/1	19	23	X			

sw2_piso2	Fa0/2	19	42		X		
sw2_piso2	Fa0/11	19	-		X		to PC11
sw2_piso2	Fa0/24	19	23			X	
sw1_piso2	Fa0/2	19	42			X	
sw1_piso2	Fa0/10	19	-		X		to PC7
sw1_piso2	Fa0/18	19	23			X	
sw1_piso2	Fa0/23	19	23	X			
sw1_piso2	Fa0/24	19	23			X	
sw2_piso1	Gig0/1	4	4	X			
sw2_piso1	Fa0/1	19	61		X		
sw2_piso1	Fa0/2	19	23			X	
sw2_piso1	Fa0/23	19	61		X		
sw2_piso1	Fa0/24	19	42		X		

table 3 - Root Bridge, port Roles and Root Port Costs (VLAN 11)

VLAN 12 (Secretariat) - Root Bridge, Port Roles and Root Port Costs

Switch	Port	PC	RPC	RP	DP	BLK	Notes
SW_DC	Gig1/0/1	4	-				
SW_DC	Gig1/0/2	4	-				
SW_DC	Gig1/0/5	19	-				to Fa port at RouterA
sw1_piso1	Gig0/1	4	4	X			
sw1_piso1	Fa0/2	19	23		X		
sw1_piso1	Fa0/10	19	-		X		to PC5
sw1_piso1	Fa0/23	19	42		X		
sw1_piso1	Fa0/24	19	42		X		
sw2_piso2	Fa0/1	19	23	X			
sw2_piso2	Fa0/2	19	42		X		
sw2_piso2	Fa0/10	19	-		X		to PC8
sw2_piso2	Fa0/24	19	23			X	
sw1_piso2	Fa0/2	19	42			X	
sw1_piso2	Fa0/18	19	23			X	
sw1_piso2	Fa0/23	19	23	X			
sw1_piso2	Fa0/24	19	23			X	
sw2_piso1	Gig0/1	4	4	X			
sw2_piso1	Fa0/1	19	61		X		
sw2_piso1	Fa0/2	19	23			X	
sw2_piso1	Fa0/23	19	61		X		
sw2_piso1	Fa0/24	19	42		X		

table 4 - Root Bridge, port Roles and Root Port Costs (VLAN 12)

VLAN 13 (Computer_science) - Root Bridge, Port Roles and Root Port Costs

Switch	Port	PC	RPC	RP	DP	BLK	Notes
SW_DC	Gig1/0/1	4	-				
SW_DC	Gig1/0/2	4	-				
SW_DC	Gig1/0/5	19	-				to Fa port at RouterA
sw1_piso1	Gig0/1	4	4	X			
sw1_piso1	Fa0/2	19	23		X		

sw1_piso1	Fa0/23	19	42		X		
sw2_piso2	Fa0/1	19	23	X			
sw2_piso2	Fa0/2	19	42		X		
sw2_piso2	Fa0/24	19	23			X	
sw1_piso2	Fa0/2	19	42			X	
sw1_piso2	Fa0/18	19	23			X	
sw1_piso2	Fa0/23	19	23	X			
sw1_piso2	Fa0/24	19	23		X		
sw2_piso1	Gig0/1	4	4	X			
sw2_piso1	Fa0/1	19	61		X		
sw2_piso1	Fa0/2	19	23			X	
sw2_piso1	Fa0/10	19	-		X		to PC6
sw2_piso1	Fa0/23	19	61		X		
sw2_piso1	Fa0/24	19	42		X		

table 5 - Root Bridge, port Roles and Root Port Costs (VLAN 13)

VLAN 99 (Native) - Root Bridge, Port Roles and Root Port Costs

Switch	Port	PC	RPC	RP	DP	BLK	Notes
SW_DC	Gig1/0/1	4	-				
SW_DC	Gig1/0/2	4	-				
SW_DC	Gig1/0/5	19	-				to Fa port at RouterA
sw1_piso1	Gig0/1	4	4	X			
sw1_piso1	Fa0/2	19	23		X		
sw1_piso1	Fa0/23	19	42		X		
sw1_piso1	Fa0/24	19	42		X		
sw2_piso2	Fa0/1	19	23	X			
sw2_piso2	Fa0/2	19	42		X		
sw2_piso2	Fa0/24	19	23			X	
sw1_piso2	Fa0/2	19	42			X	
sw1_piso2	Fa0/18	19	23			X	
sw1_piso2	Fa0/23	19	23	X			
sw1_piso2	Fa0/24	19	23			X	
sw2_piso1	Gig0/1	4	4	X			
sw2_piso1	Fa0/1	19	61		X		
sw2_piso1	Fa0/2	19	23			X	
sw2_piso1	Fa0/23	19	61		X		
sw2_piso1	Fa0/24	19	42		X		

table 6 - Root Bridge, port Roles and Root Port Costs (VLAN 99)

1.1.2 Rapid PVST

To convert to Rapid PVST we need to do in all switches:

```
SWITCH(config)# spanning-tree mode rapid-pvst
```

In the ports connected to PCs, we convert them to `port fast` and enable the `bpdu guard`

```
# sw1_piso1 -----
sw1_piso1(config)# interface fa0/10
sw1_piso1(config-if)# spanning-tree portfast
sw1_piso1(config-if)# spanning-tree bpduguard enable

# sw2_piso1 -----
sw2_piso1(config)# interface fa0/10
```

```

sw2_piso1(config-if)# spanning-tree portfast
sw2_piso1(config-if)# spanning-tree bpduguard enable

# sw1_piso2 -----
sw1_piso2(config)# interface fa0/10
sw1_piso2(config-if)# spanning-tree portfast
sw1_piso2(config-if)# spanning-tree bpduguard enable

# sw1_piso1 -----
sw1_piso1(config)# interface range fa0/10-11
sw1_piso1(config-if-range)# spanning-tree portfast
sw1_piso1(config-if-range)# spanning-tree bpduguard enable

```

SW_DC

Is the root bridge of all VLANs (11, 12, 13 and 99) with priority of 32768.

- Hello Time - 2
- Max Age - 20
- Forward Delay - 15

All three interfaces (Gig1/0/1, Gig1/0/2 and Gig1/0/5) are **Designated** for all VLANs

sw1_piso1

Is bridge of all VLANs (11, 12, 13 and 99) with priority of 32768.

- Hello Time - 2
- Max Age - 20
- Forward Delay - 15

Interface	VLAN 11	VLAN 12	VLAN 13	VLAN 99
Gig 0/1	Root	Root	Root	Root
Fa0/2	Designated	Designated	Designated	Designated
Fa0/10	-	Designated	-	-
Fa0/23	Designated	Designated	Designated	Designated
Fa0/24.	Designated	Designated	-	Designated

table 7 - sw1_piso1 port roles per VLAN

sw2_piso2

Is bridge of all VLANs (11, 12, 13 and 99) with priority of 32768.

- Hello Time - 2
- Max Age - 20
- Forward Delay - 15

Interface	VLAN 11	VLAN 12	VLAN 13	VLAN 99
Gig 0/1	Root	Root	Root	Root
Fa0/1	Designated	Designated	Designated	Designated
Fa0/2	Alternate	Alternate	Alternate	Alternate
Fa0/10	-	-	Designated	-
Fa0/23	Alternate	Alternate	Designated	Designated
Fa0/24	Designated	Designated	Designated	Designated

table 8 - sw2_piso1 port roles per VLAN

sw1_piso2

Is bridge of all VLANs (11, 12, 13 and 99) with priority of 32768.

- Hello Time - 2
- Max Age - 20
- Forward Delay - 15

Interface	VLAN 11	VLAN 12	VLAN 13	VLAN 99
Fa0/2	Alternate	Alternate	Alternate	Alternate
Fa0/10	Designated	-	-	-

Fa0/18	Designated	Designated	Alternate	Alternate
Fa0/23	Root	Root	Root	Root
Fa0/24	Alternate	Alternate	Designated	Alternate

table 9 - sw1_piso2 port roles per VLAN

sw2_piso2

Is bridge of all VLANs (11, 12, 13 and 99) with priority of 32768.

- Hello Time - 2
- Max Age - 20
- Forward Delay - 15

Interface	VLAN 11	VLAN 12	VLAN 13	VLAN 99
Fa0/1	Root	Root	Root	Root
Fa0/2	Designated	Designated	Designated	Designated
Fa0/10	-	Designated	-	-
Fa0/11	Designated	-	-	-
Fa0/24	Alternate	Alternate	Alternate	Alternate

table 10 - sw2_piso2 port roles per VLAN

1.2 Tests and Validation

1.2.1 Root bridge (RB), designated/blocked ports, and path costs (VLAN 1)

Switches / Bridges priorities for VLAN 1

Switch	Priority	MAC	Root MAC
sw1_piso2	28673	00E0.A3CE.4A46	same
SW_DC	32769	0002.4AE4.E093	00E0.A3CE.4A46
sw1_piso1	32769	0009.7C65.BA4B	00E0.A3CE.4A46
sw2_piso2	32769	00E0.8FE1.53D7	00E0.A3CE.4A46
sw2_piso1	32769	00E0.F950.631A	00E0.A3CE.4A46

table 11 - Switches / Bridges priorities (VLAN 1)

Root Bridge, Port Roles and Root Port Costs for VLAN 1

Switch	Port	PC	RPC	RP	DP	ALT/BLK
sw1_piso2	Fa0/2	19	-			
sw1_piso2	Fa0/10	19	-			
sw1_piso2	Fa0/18	19	-			
sw1_piso2	Fa0/23	19	-			
sw1_piso2	Fa0/24	19	-			
SW_DC	Gig1/0/1	4	23	X		
SW_DC	Gig1/0/2	4	23			X
SW_DC	Gig1/0/5	4	?		X	
sw1_piso1	Gig0/1	4	-		X	
sw1_piso1	Fa0/2	19	38		X	
sw1_piso1	Fa0/10	19	(PC5)		X	
sw1_piso1	Fa0/23	19	19			X
sw1_piso1	Fa0/24	19	19	X		
sw2_piso2	Fa0/1	19	38	X		
sw2_piso2	Fa0/2	19	19			X
sw2_piso2	Fa0/10	19	(PC8)		X	

sw2_piso2	Fa0/11	19	(PC9)		X	
sw2_piso2	Fa0/24	19	38			X
sw2_piso1	Gi0/1	4	27		X	
sw2_piso1	Fa0/1	19	38		X	
sw2_piso1	Fa0/2	19	38			X
sw2_piso1	Fa0/10	19	(PC6)		X	
sw2_piso1	Fa0/23	19	19	X		
sw2_piso1	Fa0/24	19	38		X	

table 12 - Port Roles and Root Port Costs (VLAN 1)

1.2.2 SW_DC as the RB (lower priority) and reassessment of blocked links (VLAN 1)

For the switch `SW_DC` to be the **Root Bridge** in all VLANs, we force his priority. Previously it was **Root Bridge** only to VLANs 11, 12 and 13, but VLAN 1 has his **Root Bridge** at `sw1_piso2`.

```
SW_DC(config) #spanning-tree vlan X priority 4096
```

1.2.3 Modification of settings between `sw1_piso1` and `sw1_piso2` to exchange blocked by forward links

In VLAN 1, to change the port roles between `sw1_piso1` and `sw1_piso2`, we *force* the cost of the *blocked* port (`sw1_piso1 - Fa0/23`) to have the value **18** instead of **19**.

```
sw1_piso1(config)#int fa0/23
sw1_piso1(config-if)#spanning-tree cost 18
```

After doing this, this port changed from `Blocked` to `Root`, and the port `sw1_piso1 - fa0/24` from `Root` to `Blocked`. This was achieved because from the point of view of the spanning tree protocol, the port with lower cost is the one that is chosen.

1.3 Practical Questions

1. What is the purpose of this command - "no ip domain-lookup"?

The `no ip domain-lookup` command in Cisco Packet Tracer is used to **disable the router or switch's default behaviour of trying to resolve any mistyped command as a hostname**.

By default, when it is typed an unrecognised word at the command line, a Cisco device assumes that we are trying to connect via Telnet to a host with that name. The device then attempts a DNS lookup to find that host's IP address. In a lab environment without a DNS server, this causes the command line to freeze for a minute or more while the device broadcasts a request that will never be answered.

Using `no ip domain-lookup` prevents this delay. After entering this command, a mistyped command results in an immediate error message, allowing the correction of the mistake without waiting

[no ip domain lookup command](#)

2. What default VLANs exist before any VLAN is configured on any equipment?

Before any configuration is done, all Cisco switches come with a set of **default VLANs** that are created automatically.

The most crucial one, which is always present, is:

1. VLAN 1

This is the **default native VLAN**.

- **Purpose:** By default, all switch ports are members of VLAN 1. All traffic traversing a trunk port (a port that carries multiple VLANs) that is not specifically tagged with a VLAN ID will be assumed to belong to the native VLAN, which is VLAN 1 by default.
- **Important Note:** For security reasons, it is a universal best practice **never to use VLAN 1 for user data traffic**. It should only be used for control plane traffic (mentioned below). Many network administrators create a different VLAN (e.g., VLAN 99) to act as the native VLAN and remove VLAN 1 from all trunks and access ports.

Other Pre-Configured VLANs

Depending on the switch model and IOS version, we might also see one or both of the following VLANs pre-created. They are used for internal switch management and communication.

- VLAN 1002 (fddi-default - Legacy FDDI network default)
- VLAN 1003 (token-ring-default - Legacy Token Ring network default)
- VLAN 1004 (fddinet-default - Legacy FDDI Net default)
- VLAN 1005 (trnet-default - Legacy Token Ring Net default)

3. What is the format of the tags introduced in Ethernet frames in trunk connections?

The format used for tagging Ethernet frames on trunk connections is defined by the **IEEE 802.1Q** standard, often called **"dot1q"**.

On an **access link** (a link connected to a normal device like a PC), frames are untagged. The switch adds or removes the VLAN information internally. On a **trunk link** (a link between switches, or to a router), frames from multiple VLANs must travel over the same physical cable. The tag is what allows the receiving switch to identify which VLAN the frame belongs to.

The 802.1Q Tag Format

The 802.1Q tag is a **4-byte (32-bit)** field that is inserted into the original Ethernet frame, right after the **Source MAC Address** field.

BEFORE 802.1Q TAGGING (Standard Ethernet Frame):

Destination MAC	Source MAC	Type/Length	Data	FCS
6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes

AFTER 802.1Q TAGGING (VLAN Tagged Frame on a Trunk):

Destination MAC	Source MAC	802.1Q TAG	Type/Length	Data	FCS
6 bytes	6 bytes	4 bytes	2 bytes	46-1500 bytes	4 bytes

4. Why is it that on a LAN that uses VLANs, on an Access-type connection the frames do not include tags?

The reason frames are **untagged on Access ports** is fundamentally about **compatibility with end-user devices**. Most end devices (like PCs, printers, IP phones, servers) are completely unaware of VLANs and expect to send and receive standard, untagged Ethernet frames.

An Access port is like a "VLAN translator" that sits between the switched network and the end device.

Objective of this design

- 1. **Simplicity and Universality:** It allows the introduction of VLANs into the network without having to upgrade or reconfigure every single PC, printer, and server to understand 802.1Q tags.
- 2. **Clear Separation of Responsibilities:**
 - **Switches** are the *VLAN-aware* devices. They handle the complexity of tagging, filtering, and trunking.
 - **End-devices** are *VLAN-unaware*. They just send and receive standard network traffic.

5. What is the tag that the ports/interfaces belonging to VLAN 1 carry?

Ports belonging to VLAN 1 do not carry a tag; they are untagged. This is because VLAN 1 is the default Native VLAN on Cisco switches.

6. When a machine receives an Ethernet frame, how does it differ if it includes the Type/Length field after the source address field or if it includes the fields associated with a VLAN?

The key difference lies in how the network interface card (NIC) and its driver **interpret the two bytes that follow the Source MAC Address**.

There are two different scenarios:

- **Scenario 1: Standard Ethernet Frame (No VLAN Tag)**

This is the original Ethernet II frame format.

Frame Structure:

Destination MAC (6)	Source MAC (6)	Type/Length (2)	Data & Padding (46-1500)	FCS (4)
---------------------	----------------	-----------------	--------------------------	---------

Processing from the receiving machine processes:

The machine looks at the **Type/Length field** (officially called the **EtherType** field when used as a type identifier). It checks the value in this 2-byte field:

- **If the value is 0x0600 (1536) or greater:** It is interpreted as an **EtherType**. This indicates the protocol of the payload contained in the "Data" field. The machine hands the frame's payload to the appropriate network protocol stack (e.g., the IPv4 stack, the IPv6 stack, etc.).
 - **Common Examples:**
 - 0x0800 = IPv4
 - 0x86DD = IPv6
 - 0x0806 = ARP
- **If the value is 0x05DC (1500) or less:** It is interpreted as the **Length** of the frame. This is used by the legacy IEEE 802.3 standard (which is less common today). The machine would then need to look at the "Data" portion to find a separate LLC (Logical Link Control) header to determine the protocol type.

Key Takeaway: In a modern standard frame, the field is an **EtherType** that directly tells the OS what's inside.

- **Scenario 2: VLAN-Tagged Frame (802.1Q)**

This is the frame after the 4-byte VLAN tag has been inserted.

Frame Structure:

Dest MAC (6)	Source MAC (6)	0x8100 (2)	TCI (2)	Type/Length (2)	Data & Padding (46-1500)	FCS (4)
--------------	----------------	------------	---------	-----------------	--------------------------	---------

Processing by the receiving machine processes:

The machine starts reading the frame as usual. After the Source MAC address, it encounters the next 2 bytes:

- i. It reads the value 0x8100 .
 - This specific value is a well-known, reserved EtherType that means "This is an 802.1Q tagged frame."
 - When the NIC sees 0x8100 , it knows that the next 4 bytes are not the payload's protocol type, but are instead a VLAN tag.
- ii. It skips the next 2 bytes (the TCI field containing Priority, CFI, and VLAN ID).
 - The machine's driver might read the VLAN ID here if it needs it for software-based VLAN processing, but for basic protocol demultiplexing, it just knows to look past this tag.
- iii. It now looks at the next 2-byte field, which is the real Type/Length field.
 - This field now contains the actual EtherType (e.g., 0x0800 for IPv4) that describes the protocol of the encapsulated payload that follows.

Side-by-Side Comparison

Step	Standard Frame (No VLAN)	VLAN-Tagged Frame (802.1Q)
1. Read Dest/Src MAC	Reads 6-byte Destination, then 6-byte Source.	Same. Reads 6-byte Destination, then 6-byte Source.
2. Interpret Next 2 Bytes	Sees value (e.g., 0x0800). This is the payload's Type.	Sees value 0x8100 . This is a signal that a VLAN tag follows.
3. Action	Sends "Data" to the protocol stack indicated by the Type (e.g., IPv4 stack).	Reads the next 4 bytes as the VLAN Tag, then moves to the next 2-byte field.
4. Find Payload Type	(Already found in step 2)	Reads the next 2-byte field. This value (e.g., 0x0800) is the real payload Type.
5. Final Action	Processes payload.	Sends "Data" to the protocol stack indicated by the Type found in step 4.

table 13 - VLAN Tag comparison

7. What are the possible consequences of passing the timers "Max Age"=20 sec and "Forward Delay"= 15 sec to half of these values?

Adjusting the STP timers to half their default values can speed up network convergence but carries significant risks. The consequences depend heavily on the network's physical size and stability.

Weighing the Consequences: Faster Convergence vs. Stability Risks

The table below summarizes the potential outcomes:

Timer Change	Potential Benefit	Risks and Consequences
Max Age: 20s → 10s	Faster detection of link failures.	Unstable Links: Ports may flap between blocking/forwarding on poor links. Topology Instability: Increased risk of transient loops if BPDUs are delayed.
Forward Delay: 15s → 7s	Drastically reduced time for ports to become active (from 30-50s to 15-25s).	Network Loops: High risk of temporary Layer 2 loops and broadcast storms. Frame Duplication: Switches may forward data frames from old topology.

table 14 - Pros and Cons of reducing timers of STP

The default values (Max Age=20, Forward Delay=15) are calculated based on a maximum network diameter of seven switches and account for BPDU propagation delays. Halving them assumes your network is smaller and has lower latency.

Understand and Tune Spanning Tree Protocol Timers

8. What is the Root Bridge (RB)? Justify.

In a network using the Spanning Tree Protocol (STP), the Root Bridge acts as the logical center or boss of the network. Its primary purpose is to serve as a common reference point that all other switches use to build a loop-free topology, allowing for redundant links without the risk of switching loops or broadcast storms.

Its key roles include:

- BPDUs Originator: The Root Bridge is the source for Bridge Protocol Data Units (BPDUs), the special frames that switches use to share STP information. These BPDUs are relayed throughout the network from the root bridge down to all other switches.
- Timer Keeper: The Root Bridge dictates critical STP timers for the entire network, including the Hello Time , Forward Delay , and Max Age .
- Topology Baseline: All other switches determine the best path to the Root Bridge.

Spanning Tree Protocol

9. By default, what is the type of active Spanning-Tree (STP) [hint: use sh span]?

Using the command `sh spanning-tree summary` it says `Switch is in pvst mode` which corresponds to **Per-VLAN Spanning Tree Plus (PVST+)**.
Using the hint command, only says `Spanning tree enabled protocol ieee`.

10. How many spanning trees are there in the implemented topology?

In the implemented topology there are 4 spanning trees, one for each VLAN. VLAN 1, VLAN 11, VLAN 12 and VLAN 13.

11. For company A, build the table for calculating the cost of the paths and determining which are the Root, Designated and Blocking ports and calculate the respective values (check in the PT which cost values are used in the calculations of the spanning trees). Are the final results you arrived at consistent with those that the PT simulator presents?

View point 1.2.1

12. What is the cost of the shortest path to Router A since PC9?

From (device - interface)	To (device - interface)	Cost
PC9 - Fa0	sw2_piso2 - Fa0/11	19
sw2_piso2 - Fa0/1	sw2_piso1 - Fa0/1	19
sw2_piso1 - Gig0/1	SW_DC - Gig1/0/2	4
SW_DC - Gig1/0/5	RouterA - Fa0/1	19

table 15 - Cost calculations from PC9 to Router A

The cost of the shortest path to Router A since PC9 is `19 + 19 + 4 + 19 = 61`

2. Enterprise A - VLAN Segmentation and Addressing

2.1 Implementation

2.1.1 VLANs

VLAN	NAME	IP GATEWAY	NETWORK	PCs
11	Accounting	172.20.11.254	172.20.11.0/24	PC7, PC9
12	Secretariat	172.20.12.254	172.20.12.0/24	PC5, PC8
13	Computer science	172.20.13.126	172.20.13.0/25	PC6

table 16 - VLANs and corresponding Networks and Gateway

Already configured, view point 1.1.1 VLANs

1.1.2 Access and Trunk Ports

Already configured, in the point 1.1.1 VLANs

2.1.3 PCs IP Assignments

PC	VLAN	IP address	Gateway
PC5	12	172.20.12.5	172.20.12.254
PC6	13	172.20.13.6	172.20.13.126
PC7	11	172.20.11.7	172.20.11.254
PC8	12	172.20.12.8	172.20.12.254
PC9	11	172.20.11.9	172.20.11.254

table 17 - PCs IP address assignments

We decided, that the last octet of the assigned IP address corresponds to the PC number

2.2 Test and Validation

2.2.1 Intra-VLAN and Inter-VLAN connectivity

With the configuration that we made, we have connectivity intra-VLAN but not inter-VLAN. To achieve inter-VLAN connectivity we need to configure the Router A with `Router-in-a-Stick`, switches cannot change the VLAN tag, so we have not connectivity inter-VLANs

2.2.2 Connectivity between the PCs on each VLAN

SW_DC `show vlan brief` and `show interfaces trunk`

```
SW_DC#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/6, Gig1/0/7 Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11 Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15 Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19 Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23 Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
11	Accounting	active	
12	Secretariat	active	
13	Computer_science	active	
99	VLAN0099	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW_DC#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig1/0/1	on	802.1q	trunking	99
Gig1/0/2	on	802.1q	trunking	99
Gig1/0/5	on	802.1q	trunking	99

Port	Vlans allowed on trunk
Gig1/0/1	11-13
Gig1/0/2	11-13
Gig1/0/5	11-13

```

Port      Vlans allowed and active in management domain
Gig1/0/1  11,12,13
Gig1/0/2  11,12,13
Gig1/0/5  11,12,13

Port      Vlans in spanning tree forwarding state and not pruned
Gig1/0/1  11,12,13
Gig1/0/2  11,12,13
Gig1/0/5  11,12,13

```

sw1_piso1 show vlan brief and show interfaces trunk

```
sw1_piso1#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Gig0/2
11	Accounting	active	
12	Secretariat	active	Fa0/10
13	Computer_science	active	
99	VLAN0099	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
sw1_piso1#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	on	802.1q	trunking	99
Fa0/23	on	802.1q	trunking	99
Fa0/24	on	802.1q	trunking	99
Gig0/1	on	802.1q	trunking	99

```

Port      Vlans allowed on trunk
Fa0/2     11-13
Fa0/23    11-13
Fa0/24    11-13
Gig0/1    11-13

```

```

Port      Vlans allowed and active in management domain
Fa0/2     11,12,13
Fa0/23    11,12,13
Fa0/24    11,12,13
Gig0/1    11,12,13

```

```

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/2     11,12,13
Fa0/23    11,12,13
Fa0/24    11,12,13
Gig0/1    11,12,13

```

sw2_piso1 show vlan brief and show interfaces trunk

```
sw1_piso2#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Gig0/1 Gig0/2
11	Accounting	active	Fa0/10
12	Secretariat	active	
13	Computer_science	active	
99	VLAN0099	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
sw1_piso2#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	on	802.1q	trunking	99
Fa0/18	on	802.1q	trunking	99
Fa0/23	on	802.1q	trunking	99
Fa0/24	on	802.1q	trunking	99

Port	Vlans allowed on trunk
Fa0/2	11-13
Fa0/18	11-13
Fa0/23	11-13
Fa0/24	11-13

Port	Vlans allowed and active in management domain
Fa0/2	11,12,13
Fa0/18	11,12,13
Fa0/23	11,12,13
Fa0/24	11,12,13

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/2	none
Fa0/18	none
Fa0/23	11,12,13
Fa0/24	none

sw2_piso2 show vlan brief and show interfaces trunk

```
sw2_piso2#sh vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Gig0/1 Gig0/2
11 Accounting	active	Fa0/11
12 Secretariat	active	Fa0/10
13 Computer_science	active	
99 VLAN0099	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	


```
sw2_piso2#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	99
Fa0/2	on	802.1q	trunking	99
Fa0/24	on	802.1q	trunking	99

Port	Vlans allowed on trunk
Fa0/1	11-13
Fa0/2	11-13
Fa0/24	11-13

Port	Vlans allowed and active in management domain
Fa0/1	11,12,13
Fa0/2	11,12,13
Fa0/24	11,12,13

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	11,12,13
Fa0/2	11,12,13
Fa0/24	none

VLAN 11 - Accounting connectivity between PC7 and PC9

```
C:\>ipconfig
```

FastEthernet0 Connection:(default port)

```

Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: FE80::201:43FF:FE2A:258E
IPv6 Address.....: ::
IPv4 Address.....: 172.20.11.7
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        172.20.11.254

```

```
C:\>ping 172.20.11.9
```

Pinging 172.20.11.9 with 32 bytes of data:

```

Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128

```

Ping statistics for 172.20.11.9:


```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

VLAN 12 - Accounting connectivity between PC5 and PC8

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::230:A3FF:FEB5:B871
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 172.20.12.5
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                172.20.12.254

C:\>ping 172.20.12.8

Pinging 172.20.12.8 with 32 bytes of data:

Reply from 172.20.12.8: bytes=32 time<1ms TTL=128
Reply from 172.20.12.8: bytes=32 time<1ms TTL=128
Reply from 172.20.12.8: bytes=32 time<1ms TTL=128
Reply from 172.20.12.8: bytes=32 time<1ms TTL=128

Ping statistics for 172.20.12.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

VLAN 13 (Computer_science) has only one PC connected, PC6

2.3 Practical Questions

1. What is the cost of the shortest path to Router A since PC9?

From (device - interface)	To (device - interface)	Cost
PC9 - Fa0	sw2_piso2 - Fa0/11	19
sw2_piso2 - Fa0/1	sw2_piso1 - Fa0/1	19
sw2_piso1 - Gig0/1	SW_DC - Gig1/0/2	4
SW_DC - Gig1/0/5	RouterA - Fa0/1	19

table 18 - Costs from PC9 to Router A

The cost of the shortest path to Router A since PC9 is $19 + 19 + 4 + 19 = 61$

2. Explain what actions were required to accommodate the topology requirements.

To accommodate this topology requirements we need to configure, the VLANs, trunk and access interfaces on the switches, and configure VLAN IP addresses on SW_DC (VLAN gateways)

3. Did you achieve inter vlan connectivity in this phase? Explain the observed behavior.

No. Because switches cannot change VLAN tags, we cannot communicate between PCs that are connected in distinct VLANs. To achieve that we need to configure a Router with Router-on-a-Stick.

3. Enterprise A — Router-on-a-Stick (RoS) & L3 Rules (no ACLs)

3.1 Implementation

3.1.1 Creation and configuration of sub interfaces to implement the L3 (Rules without ACLs)

- Accounting & Secretariat: no access to any other internal or external VLAN/network
- Computer_science: access to Secretariat and outside Company A

Rule 1: Accounting & Secretariat Isolation

- To achieve this we didn't create routes between VLAN 11 and VLAN 12
- These VLANs remain isolated by default since there's no routing between them

Rule 2: Computer_science Access

- Computer_science (VLAN 13) can access Secretariat (VLAN 12) because both have routes through the router
- For "outside Company A" access, we will need to configure routing toward the ISP in a later phases

Configuration of default Gateways on PCs

- PC7 & PC9 (Accounting): Default gateway = 172.20.11.254
- PC5 & PC8 (Secretariat): Default gateway = 172.20.12.254
- PC6 (Computer_science): Default gateway = 172.20.13.126

3.1.2 Trunk link between Router A and SW_DC

To configure a trunk link between Router A and SW_DC, we made the next configuration on Router A

```
RouterA(config)# interface fa0/1
RouterA(config-if)# description Trunk_to_SW_DC
RouterA(config-if)# no shutdown
RouterA(config-if)# no ip address
```

3.1.3 Sub interfaces with encapsulation dot1Q and IPs (parent interface no IP)

```
# VLAN 11 Accounting -----
RouterA(config)# interface FastEthernet0/1.11
RouterA(config-subif)# description VLAN 11
RouterA(config-subif)# encapsulation dot1Q 11
RouterA(config-subif)# ip address 172.20.11.254 255.255.255.0
RouterA(config-subif)# no shutdown
RouterA(config-subif)# exit

# VLAN 12 Secretariat -----
RouterA(config)# interface FastEthernet0/1.12
RouterA(config-subif)# description VLAN 12
RouterA(config-subif)# encapsulation dot1Q 12
RouterA(config-subif)# ip address 172.20.12.254 255.255.255.0
RouterA(config-subif)# shutdown
RouterA(config-if)# exit

# VLAN 13 Computer_science -----
RouterA(config)# interface FastEthernet0/1.13
RouterA(config-subif)# description VLAN 13
RouterA(config-subif)# encapsulation dot1Q 13
RouterA(config-subif)# ip address 172.20.13.126 255.255.255.128
RouterA(config-subif)# no shutdown
RouterA(config-subif)# exit
```

3.1.4 Name routers/switches

To name router/switches we used the `hostname` command:

```
Router(config)# hostname RouterN
RouterN(config)#
```

3.1.5 Router / Switches initial message

To set up an initial message for those who enter the machines, we use the command:

```
RouterN (config)# banner login ^C
--- Router N ---
-----
--- UNAUTHORIZED ACCESS IS PROHIBITED ---
--- Entries not authorized by lei ---
--- (Law 109/2009 of 15 September) ---
```

3.2 Test and Validation

You need to test connectivity from multiple points to prove both **allowed access** and **enforced restrictions**.

Communications from and to VLAN 11 (Accounting)

```
# From PC7 - VLAN 11 -----
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::201:43FF:FE2A:258E
    IPv6 Address.....: ::
    IPv4 Address.....: 172.20.11.7
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                           172.20.11.254

# To PC9 - VLAN 11 -----
C:\>ping 172.20.11.9

Pinging 172.20.11.9 with 32 bytes of data:

Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128
Reply from 172.20.11.9: bytes=32 time<1ms TTL=128

Ping statistics for 172.20.11.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

# To PC5 - VLAN 12 -----
C:\>ping 172.20.12.5

Pinging 172.20.12.5 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.20.12.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

# To PC6 - VLAN 13 -----
C:\>ping 172.20.13.6

Pinging 172.20.13.6 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.20.13.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Communications from and to VLAN 12 (Secretariat)

```
# From PC5 - VLAN 12 -----
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::230:A3FF:FEB5:B871
    IPv6 Address.....: ::
    IPv4 Address.....: 172.20.12.5
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                           172.20.12.254

# To PC6 - VLAN 13 -----
C:\>ping 172.20.13.6

Pinging 172.20.13.6 with 32 bytes of data:

Reply from 172.20.13.6: bytes=32 time=16ms TTL=127
Reply from 172.20.13.6: bytes=32 time<1ms TTL=127
```

```

Reply from 172.20.13.6: bytes=32 time<1ms TTL=127
Reply from 172.20.13.6: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.13.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

# To PC7 - VLAN 11 -----
C:\>ping 172.20.11.7

Pinging 172.20.11.7 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.20.11.7:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

# To PC8 - VLAN 12 -----
C:\>ping 172.20.12.8

Pinging 172.20.12.8 with 32 bytes of data:

Reply from 172.20.12.8: bytes=32 time<1ms TTL=128
Reply from 172.20.12.8: bytes=32 time=12ms TTL=128
Reply from 172.20.12.8: bytes=32 time<1ms TTL=128
Reply from 172.20.12.8: bytes=32 time<1ms TTL=128

Ping statistics for 172.20.12.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

```

Communications from and to VLAN 13 (Computer Science)

```

# From PC6 - VLAN 13 -----
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::290:CFF:FEC9:2A2
    IPv6 Address.....: ::
    IPv4 Address.....: 172.20.13.6
    Subnet Mask.....: 255.255.255.128
    Default Gateway.....: ::
                        172.20.13.127

# To PC8 - VLAN 12 -----
C:\>ping 172.20.12.8

Pinging 172.20.12.8 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 172.20.12.8: bytes=32 time=1ms TTL=127
Reply from 172.20.12.8: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.12.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

# To PC9 - VLAN 11 -----
C:\>ping 172.20.11.9

Pinging 172.20.11.9 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.20.11.9:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Router-Level Validation Commands

Router A configuration:

Show subinterface status:

```
RouterA# show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.20.1.2       YES NVRAM  up          up
FastEthernet0/1  unassigned      YES NVRAM  up          up
FastEthernet0/1.11 172.20.11.254  YES manual administratively down down
FastEthernet0/1.12 172.20.12.254  YES manual up          up
FastEthernet0/1.13 172.20.13.126  YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
```

Show Routing Table:

```
RouterA# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is 10.20.1.1 to network 0.0.0.0

```

10.0.0.0/30 is subnetted, 1 subnets
C      10.20.1.0 is directly connected, FastEthernet0/0
172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.20.12.0/24 is directly connected, FastEthernet0/1.12
C      172.20.13.0/25 is directly connected, FastEthernet0/1.13
S*    0.0.0.0/0 [1/0] via 10.20.1.1
```

Switch-Level Validation

SW_DC interfaces trunk:

```
SW_DC#sh interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig1/0/1  on        802.1q         trunking    99
Gig1/0/2  on        802.1q         trunking    99
Gig1/0/5  on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig1/0/1  11-13
Gig1/0/2  11-13
Gig1/0/5  11-13

Port      Vlans allowed and active in management domain
Gig1/0/1  11,12,13
Gig1/0/2  11,12,13
Gig1/0/5  11,12,13

Port      Vlans in spanning tree forwarding state and not pruned
Gig1/0/1  11,12,13
Gig1/0/2  11,12,13
Gig1/0/5  11,12,13
```

Advanced Validation with Traceroute (Tracert)

```
# From PC5 - VLAN 12 -----
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: FE80::230:A3FF:FEB5:B871
IPv6 Address.....: ::
IPv4 Address.....: 172.20.12.5
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        172.20.12.254

To PC6 - VLAN 13 -----
C:\>tracert 172.20.13.6

Tracing route to 172.20.13.6 over a maximum of 30 hops:

  1  *      *      *      Request timed out.
  2  *      *      *      Request timed out.
  3  *      *      *      Request timed out.
  .....
 29  *      *      *      Request timed out.
 30  *      *      *      Request timed out.

Trace complete.
```

```
# To PC7 - VLAN 11 -----
C:\>tracert 172.20.11.7

Tracing route to 172.20.11.7 over a maximum of 30 hops:

  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
    .....
 29  *          *          *          Request timed out.
 30  *          *          *          Request timed out.

Trace complete.
```

```
# To PC8 - VLAN 12 -----
C:\>tracert 172.20.12.8

Tracing route to 172.20.12.8 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      172.20.12.8

Trace complete.
```

```
# From PC6 - VLAN 13 -----
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::290:CFF:FEC9:2A2
    IPv6 Address.....: ::
    IPv4 Address.....: 172.20.13.6
    Subnet Mask.....: 255.255.255.128
    Default Gateway.....: ::
                        172.20.13.127

# To PC9 - VLAN 11 -----
C:\>tracert 172.20.11.9

Tracing route to 172.20.11.9 over a maximum of 30 hops:

  1  *          0 ms      0 ms      172.20.13.126
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
    .....
 29  *          *          *          Request timed out.
 30  *          *          *          Request timed out.

# To PC8 - VLAN 12 -----
C:\>tracert 172.20.12.8

Tracing route to 172.20.12.8 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      172.20.13.126
  2  *          0 ms      0 ms      172.20.12.8

Trace complete.
```

3.3 Practical Questions

1. Explain the advantages and disadvantages of the Router-on-a-stick functionality

Advantages of Router-on-a-Stick

1. Cost-Effectiveness:

- This is the primary advantage. It conserves physical router interfaces. High-speed router interfaces can be very expensive. By using a single interface to handle routing for many VLANs, you save significant hardware costs.

2. Simplified Cabling:

- Only a single physical cable is required between the router and the switch. This reduces cable clutter and simplifies the physical network layout.

3. Scalability for VLANs:

- It's very easy to add a new VLAN. You simply create a new sub interface on the router and configure the switch accordingly. There's no need to install a new physical module or card.

4. Efficient Use of Hardware:

- It allows you to leverage the routing capability of a router you already own without requiring a more advanced (and expensive) layer 3 switch.

5. Clear Separation of Duties:

- It enforces a clear network design: the switch handles layer 2 VLAN segmentation and trunking, while the router handles layer 3 inter-VLAN routing.

Disadvantages of Router-on-a-Stick

1. Single Point of Failure:

- The single physical link and the single router interface become critical points of failure. If either fails, *all* inter-VLAN communication is severed.

2. Performance and Latency Bottleneck:

- This is the most significant disadvantage. **All inter-VLAN traffic must pass through this single interface.**
- The throughput is limited by the speed of that one interface (e.g., 1 Gbps). If VLANs are communicating heavily, this link can become saturated, causing network congestion and high latency.
- The router's CPU has to process every single packet moving between VLANs, which can be taxing on older or less powerful routers.

3. Limited by Router Processing Power:

- Unlike a Layer 3 switch, which uses specialized hardware (ASICs) for high-speed switching, a router uses its general-purpose CPU for routing. This process is much slower, adding more latency.

4. Increased Complexity in Configuration:

- While the cabling is simple, the configuration is more complex than on a Layer 3 switch. You must configure sub interfaces, 802.1Q tagging, and ensure the switch trunk is set up correctly. A misconfiguration on one sub interface can affect multiple VLANs.

5. Potential for Congestion:

- Traffic has to make **two trips** across the same link for a single communication flow (in to the router, then back out to the destination VLAN). This "hairpinning" effect is inherently less efficient.

2. Explain how you enforced the three communication rules without ACLs

To isolate VLAN 12 (Secretariat) from the rest of the network, in the `Router A`, we shutdown the sub interface fa0/1.12.

To make VLAN 13 (Computer Science) communicate with VLAN 11 (Accounting), the other two sub interfaces must be `no shutdown`. But this way, without `ACLs` we cannot block VLAN 11 (Accounting) to communicate with VLAN 13 (Computer Science)

3. Provide short command outputs proving each rule is satisfied/blocked as required

View 3.2

4. Enterprise B — Segmentation & ISP L2 Interconnection

4.1 Implementation

4.1.1 Company B VLANs and IP Assignments

Remove previous configuration

```
SwEmpresaB(config)#no vlan 20
SwEmpresaB(config)#no vlan 40
```

Change native VLAN from VLAN 1 to VLAN 99

```
SwEmpresaB(config)#vlan 99
SwEmpresaB(config-vlan)#interface fa0/1
SwEmpresaB(config-if)#switchport mode trunk
SwEmpresaB(config-if)#switchport trunk native vlan 99
SwEmpresaB(config-if)#switchport trunk allowed vlan 99
SwEmpresaB(config-if)#switchport nonegotiate
```

Add VLAN 2 (Servers) and VLAN 3 (Engineering)

```
SwEmpresaB(config)#vlan 2
SwEmpresaB(config-vlan)#name Servers
SwEmpresaB(config-vlan)#vlan 3
SwEmpresaB(config-vlan)#name Engineering

SwEmpresaB(config-vlan)#interface fa0/1
SwEmpresaB(config-if)#switchport mode trunk
SwEmpresaB(config-if)#switchport trunk allowed vlan add 2,3
SwEmpresaB(config-if)#switchport nonegotiate
```

Configure access ports to end-devices

```
SwEmpresaB(config)#int fa0/10
SwEmpresaB(config-if)#switchport mode access
SwEmpresaB(config-if)#switchport access vlan 2
SwEmpresaB(config-if)#switchport nonegotiate

SwEmpresaB(config-if)#int fa0/11
SwEmpresaB(config-if)#switchport mode access
SwEmpresaB(config-if)#switchport access vlan 3
SwEmpresaB(config-if)#switchport nonegotiate

SwEmpresaB(config-if)#int fa0/12
SwEmpresaB(config-if)#switchport mode access
SwEmpresaB(config-if)#switchport access vlan 3
SwEmpresaB(config-if)#switchport nonegotiate
```

At router, we made a configuration *router-in-a-stick* and assigned IP addresses to sub interfaces

```
# Remove previous configurations -----
RouterB(config)#no int fa0/1.20
RouterB(config)#no int fa0/1.40

# New configurations -----
RouterB(config)#int fa0/1.2
RouterB(config-subif)#description VLAN 2 Servers
RouterB(config-subif)#encapsulation dot1Q 2
RouterB(config-subif)#ip address 192.168.1.30 255.255.255.224

RouterB(config-subif)#int fa0/1.3
RouterB(config-subif)#description VLAN 3 Engineering
RouterB(config-subif)#encapsulation dot1Q 3
RouterB(config-subif)#ip address 192.168.2.30 255.255.255.224
```

4.1.2 Implementation of the ISP topology for interconnection with customers

In all the router we turned of RIP with the command:

```
Router(config)#no router rip
```

Router 1 - to Company A

```
interface FastEthernet1/0
ip address 10.20.1.1 255.255.255.252
duplex auto
speed auto
```


Router 3 - to Company B

```
interface FastEthernet1/0
ip address 10.20.1.5 255.255.255.252
duplex auto
speed auto
```

4.1.3 Building of VLAN paths in the switch fabric

The switch fabric was already configured, using PVST, with VLAN 90 (Company A) and VLAN 95 (Company B) with layer 2 redundancy for trunk ports and access ports, as shown in the tables below:

Swaccesso-A

Interface	Mode	VLAN	connected to
Fa0/1	trunk	90, 95	swdistribution-1
Fa0/2	access	90	RouterA
Fa0/24	trunk	90, 95	swaccesso-B

table 19 - Interfaces configuration for Swaccesso-A

Swaccesso-B

Interface	Mode	VLAN	connected to
Fa0/1	trunk	90, 95	swdistribution-2
Fa0/2	access	95	RouterB
Fa0/24	trunk	90, 95	swaccesso-A

table 20 - Interfaces configuration for Swaccesso-B

Sdistribution-1

Interface	Mode	VLAN	connected to
Fa0/1	access	90	Router1
Fa0/2	trunk	90, 95	swaccesso-A
Fa0/24	trunk	90, 95	swdistribution-2

table 21 - Interfaces configuration for Sdistribution-1

Sdistribution-2

Interface	Mode	VLAN	Connected to
Fa0/1	access	95	Router3
Fa0/2	trunk	90, 95	swaccesso-B
Fa0/24	trunk	90, 95	swdistribution-1

table 22 - Interfaces configuration for Sdistribution-2

4.1.4 Assignment of IP address to Router 1, Router 3, Router A, and Router B

Interfaces already correctly configured in the .pkt file:

Router	Interface	IP address	Connected to
Router1	Fa1/0	10.20.1.1/30	Swdistribution-1
Router3	Fa1/0	10.20.2.5/30	Swdistribution-2
RouterA	Fa0/0	10.20.1.2/30	Swaccesso-A
RouterB	Fa0/0	10.20.1.6/30	Swaccesso-B

table 23 - Interfaces configuration for Routers 1, 2, A and B

4.2 Test and Validation

4.2.1 End-to-end L3 connectivity: R1 <-> RA and R2 <-> RB

Connectivity: R1 <-> RA

```
Router1>ping 10.20.1.2
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/12 ms
```

Connectivity: R2 <-> RB

```
Router3>ping 10.20.1.6
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

4.3 Practical Questions

1. Count STP trees; list blocked ports in VLANs 90 and 95 and justify.

Counting with default VLAN 1, we have 4 VLANs and there are no blocked ports because `fa0/1` is trunk for all VLANs and the remaining are `access` ports connected to end devices.

```
SwEmpresaB# sh spanning-tree summary
Switch is in pvst mode
Root bridge for: Servers Engineering VLAN0099
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
PortFast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0002	0	0	2	0	2
VLAN0003	0	0	3	0	3
VLAN0099	0	0	1	0	1

4 vlans	0	0	6	0	6

2. State the advantage/goal of trunk pruning in the ISP fabric.

Enhanced Security

Primary Goal: Prevent unauthorised VLAN hopping and cross-company communication

In our topology:

- **VLAN 90** carries Company A's traffic
- **VLAN 95** carries Company B's traffic
- **VLAN 99** is the native VLAN for management

Without pruning: If a trunk accidentally carries all VLANs (1-4094), there's a risk of:

- Company A potentially accessing Company B's traffic (and vice versa)
- VLAN hopping attacks between customer networks

With pruning: Each trunk only carries exactly the VLANs needed for its specific path.

Reduced Broadcast/Multicast Traffic

Goal: Minimise unnecessary network overhead

- Broadcasts from Company A (VLAN 90) should NOT flood to Company B's switches
- Multicast traffic stays contained within its intended path
- Reduces CPU load on all switches in the fabric

Optimized Bandwidth Utilization

Goal: Preserve trunk capacity for legitimate traffic

- Prevents Company A's traffic from consuming bandwidth on links dedicated to Company B
- Each customer's traffic only uses the specific path designed for them

Prevention of STP Issues

Goal: Maintain stable and predictable spanning tree behaviour

- Without pruning, all VLANs would participate in STP calculations across all trunks
- This can lead to:
 - Unnecessary STP reconvergence events

- Suboptimal root bridge elections
- Potential bridging loops involving unnecessary VLANs

Improved Troubleshooting and Management

Goal: Simplify network operations

- Clear, predictable VLAN paths make problem isolation easier
- Network maps accurately reflect actual traffic flows
- Reduces configuration complexity

Applying Trunk Pruning to Your Specific Topology

The pruning was implemented using:

Swacesso-A

```
# Should only carry VLANs 90, 95 (not VLAN 99 or others)
interface FastEthernet0/1
  switchport trunk allowed vlan 90,95

interface FastEthernet0/24
  switchport trunk allowed vlan 90,95
```

Swacesso-B

```
# Should only carry VLANs 90, 95
interface FastEthernet0/1
  switchport trunk allowed vlan 90,95

interface FastEthernet0/24
  switchport trunk allowed vlan 90,95
```

Swdistribution-1

```
# Fa0/1 (to Router1) - access port for VLAN 90 (correct)
# Fa0/2 (to Swacesso-A) - should only carry VLANs 90,95
interface FastEthernet0/2
  switchport trunk allowed vlan 90,95

# Fa0/24 (to Swdistribution-2) - should only carry VLANs 90,95
interface FastEthernet0/24
  switchport trunk allowed vlan 90,95
```

Swdistribution-2

```
# Fa0/1 (to Router3) - access port for VLAN 95 (correct)
# Fa0/2 (to Swacesso-B) - should only carry VLANs 90,95
interface FastEthernet0/2
  switchport trunk allowed vlan 90,95

# Fa0/24 (to Swdistribution-1) - should only carry VLANs 90,95
interface FastEthernet0/24
  switchport trunk allowed vlan 90,95
```

The primary goal of trunk pruning in your ISP fabric is to maintain strict isolation between Company A and Company B while optimising network performance and security. It ensures that each customer's traffic follows its designated path without interference or security risks from other customers' VLANs.

3. Explain the chosen RB priorities and observed blocked ports.

The RB for VLAN 90 (Company A) is the `Swdistribution-1`, in the same way, the RB for VLAN 95 (Company B) is the `Swdistribution-2`. This configuration ensures optimal and predictable data paths.

There is only a blocked (alternate) port between `Swacesso-A` and `Swacesso-B` to prevent a loop, as it's the least optimal path to the root bridge for one of them.

5. Static Routing → OSPF Core & “Internet” Loopback → Public Addressing Test

5.1 Implementation

5.1.1 Static routing (initial)

5.1.1.1 Default static routes on company routers toward the ISP; identify next-hop and purpose.

Company A (RouterA) - Toward ISP Router1

Next-Hop Identification:

- **RouterA's Interface:** FastEthernet0/0
- **ISP Router1's Interface:** FastEthernet1/0 with IP 10.20.1.1
- **Next-Hop IP:** 10.20.1.1 (Router1's interface facing RouterA)

Configuration:

```
RouterA(config)# ip route 0.0.0.0 0.0.0.0 10.20.1.1
```

Company B (RouterB) - Toward ISP Router3

Next-Hop Identification:

- **RouterB's Interface:** FastEthernet0/0
- **ISP Router3's Interface:** FastEthernet1/0 with IP 10.20.1.5
- **Next-Hop IP:** 10.20.1.5 (Router3's interface facing RouterB)

Configuration:

```
RouterB(config)# ip route 0.0.0.0 0.0.0.0 10.20.1.5
```

5.1.1.2 Static routing on R1–R4, add required static routes to reach company blocks.

Router1 Configuration:

```
# IP Assignments -----
Router1(config)# int fa0/0
Router1(config-if)# ip address 10.12.10.1 255.255.255.252
Router1(config-if)# int fa0/1
Router1(config-if)# ip address 10.14.10.1 255.255.255.252

# Static Routes -----
Router1(config)# ip route 0.0.0.0 0.0.0.0 10.12.10.2 # Gateway last resort
Router1(config)# ip route 172.20.11.0 255.255.255.0 10.20.1.2 # Company A (VLAN 11)
Router1(config)# ip route 172.20.12.0 255.255.254.0 10.20.1.2 # Company A (VLAN 12, 13)
Router1(config)# ip route 192.168.1.0 255.255.255.224 10.14.10.2 # Company B (VLAN 2)
Router1(config)# ip route 192.168.2.0 255.255.255.224 10.14.10.2 # Company B (VLAN 3)
```

Router2 Configuration:

```
# IP Assignments -----
Router2(config)# int fa0/0
Router2(config-if)# ip address 10.12.10.2 255.255.255.252
Router2(config-if)# int fa0/1
Router2(config-if)# ip address 10.23.10.1 255.255.255.252

# Static Routes -----
Router2(config)# ip route 0.0.0.0 0.0.0.0 Loopback0 # Gateway last resort
Router2(config)# ip route 10.20.1.0 255.255.255.252 10.12.10.1 # To Router 1
Router2(config)# ip route 10.20.1.4 255.255.255.252 10.23.10.2 # To Router 3
Router2(config)# ip route 172.20.11.0 255.255.255.0 10.12.10.1 # To Company A (VLAN 11)
Router2(config)# ip route 172.20.12.0 255.255.254.0 10.12.10.1 # To Company A (VLAN 12, 13)
Router2(config)# ip route 192.168.1.0 255.255.255.224 10.23.10.2 # To Company B (VLAN 2)
Router2(config)# ip route 192.168.2.0 255.255.255.224 10.23.10.2 # To Company B (VLAN 3)
```

Router3 Configuration:

```
# IP Assignments -----
Router3(config)# int fa0/0
Router3(config-if)# ip address 10.34.10.1 255.255.255.252
Router3(config-if)# int fa0/1
Router3(config-if)# ip address 10.23.10.2 255.255.255.252

# Static Routes -----
Router3(config)# ip route 0.0.0.0 0.0.0.0 10.23.10.1 # Gateway last resort
Router3(config)# ip route 10.20.1.0 255.255.255.252 10.14.10.1 # To Router 1
```

```

Router3(config)# ip route 172.20.11.0 255.255.255.0 10.34.10.2 # To Company A (VLAN 11)
Router3(config)# ip route 172.20.12.0 255.255.254.0 10.34.10.2 # To Company A (VLAN 12, 13)
Router3(config)# ip route 192.168.1.0 255.255.255.224 10.20.1.6 # To Company B (VLAN 2)
Router3(config)# ip route 192.168.2.0 255.255.255.224 10.20.1.6 # To Company B (VLAN 3)

```

Router4 Configuration:

```

# IP Assignments -----
Router4(config)# int fa0/0
Router4(config-if)# ip address 10.14.10.2 255.255.255.252
Router4(config-if)# int fa0/1
Router4(config-if)# ip address 10.34.10.2 255.255.255.252

Router4(config)# ip route 0.0.0.0 0.0.0.0 10.14.10.1 # Gateway last resort
Router4(config)# ip route 10.20.1.0 255.255.255.252 10.14.10.1 # To Router 1
Router4(config)# ip route 10.20.1.4 255.255.255.252 10.34.10.1 # To Router 3
Router4(config)# ip route 172.20.11.0 255.255.255.0 10.14.10.1 # To Company A (VLAN 11)
Router4(config)# ip route 172.20.12.0 255.255.254.0 10.14.10.1 # To Company A (VLAN 12, 13)
Router4(config)# ip route 192.168.1.0 255.255.255.224 10.34.10.1 # To Company B (VLAN 2)
Router4(config)# ip route 192.168.2.0 255.255.255.224 10.34.10.1 # To Company B (VLAN 3)

```

Routing tables:

```

Router1 -----
Gateway of last resort is 10.12.10.2 to network 0.0.0.0

    10.0.0.0/30 is subnetted, 3 subnets
C       10.12.10.0 is directly connected, FastEthernet0/0
C       10.14.10.0 is directly connected, FastEthernet0/1
C       10.20.1.0 is directly connected, FastEthernet1/0
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.20.11.0/24 [1/0] via 10.20.1.2
S       172.20.12.0/23 [1/0] via 10.20.1.2
    192.168.1.0/27 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 10.14.10.2
    192.168.2.0/27 is subnetted, 1 subnets
S       192.168.2.0 [1/0] via 10.14.10.2
S*    0.0.0.0/0 [1/0] via 10.12.10.2

Router2 -----
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
    10.0.0.0/30 is subnetted, 4 subnets
C       10.12.10.0 is directly connected, FastEthernet0/0
S       10.20.1.0 [1/0] via 10.12.10.1
S       10.20.1.4 [1/0] via 10.23.10.2
C       10.23.10.0 is directly connected, FastEthernet0/1
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.20.11.0/24 [1/0] via 10.12.10.1
S       172.20.12.0/23 [1/0] via 10.12.10.1
    192.168.1.0/27 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 10.23.10.2
    192.168.2.0/27 is subnetted, 1 subnets
S       192.168.2.0 [1/0] via 10.23.10.2
S*    0.0.0.0/0 is directly connected, Loopback0

Router3 -----
Gateway of last resort is 10.23.10.1 to network 0.0.0.0

    10.0.0.0/30 is subnetted, 3 subnets
C       10.20.1.4 is directly connected, FastEthernet1/0
C       10.23.10.0 is directly connected, FastEthernet0/1
C       10.34.10.0 is directly connected, FastEthernet0/0
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.20.11.0/24 [1/0] via 10.34.10.2
S       172.20.12.0/23 [1/0] via 10.34.10.2
    192.168.1.0/27 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 10.20.1.6
    192.168.2.0/27 is subnetted, 1 subnets
S       192.168.2.0 [1/0] via 10.20.1.6
S*    0.0.0.0/0 [1/0] via 10.23.10.1

Router4 -----
Gateway of last resort is 10.14.10.1 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
    10.0.0.0/30 is subnetted, 4 subnets
C       10.14.10.0 is directly connected, FastEthernet0/0
S       10.20.1.0 [1/0] via 10.14.10.1
S       10.20.1.4 [1/0] via 10.34.10.1
C       10.34.10.0 is directly connected, FastEthernet0/1
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks

```

```
S      172.20.11.0/24 [1/0] via 10.14.10.1
S      172.20.12.0/23 [1/0] via 10.14.10.1
      192.168.1.0/27 is subnetted, 1 subnets
S      192.168.1.0 [1/0] via 10.34.10.1
      192.168.2.0/27 is subnetted, 1 subnets
S      192.168.2.0 [1/0] via 10.34.10.1
S*    0.0.0.0/0 [1/0] via 10.14.10.1
```

5.1.1.3 Verification of which pings succeed before OSPF and why.

```
# From Router2, connectivity test to Company A and B -----
Router2# ping 192.168.2.2   # Company B - PC2
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router2# ping 172.20.13.6   # Company A - PC6
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

# From Router4, connectivity test to Company A and B -----
Router4# ping 192.168.2.1   # Company B - PC1
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/15 ms
Router4# ping 172.20.12.8   # Company A - PC8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/8/34 ms

# Test loopback connectivity. -----
Router1# ping 8.8.8.8       # R2's loopback
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/8/34 ms
Router3# ping 8.8.8.8       # R2's loopback
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/8/34 ms
```

5.1.2 OSPF (ISP core)

5.1.2.1 Configuration of OSPF on R1–R4 (single area) per design

Router1 Configuration:

```
Router1(config)# router ospf 1
Router1(config-router)# router-id 1.1.1.1
Router1(config-router)# network 10.12.10.0 0.0.0.3 area 0   # Link para R2
Router1(config-router)# network 10.14.10.0 0.0.0.3 area 0   # Link para R4
Router1(config-router)# network 10.20.1.0 0.0.0.3 area 0     # Company A
Router1(config-router)# passive-interface default
Router1(config-router)# no passive-interface FastEthernet0/0 # Para R2
Router1(config-router)# no passive-interface FastEthernet0/1 # Para R4
```

Router2 Configuration:

```
Router2(config)# router ospf 1
Router2(config-router)# router-id 2.2.2.2
Router2(config-router)# network 10.12.10.0 0.0.0.3 area 0   # Link para R1
Router2(config-router)# network 10.23.10.0 0.0.0.3 area 0   # Link para R3
Router2(config-router)# network 8.8.8.8 0.0.0.0 area 0       # Loopback0 (Internet)
Router2(config-router)# passive-interface default
Router2(config-router)# no passive-interface FastEthernet0/0 # Para R1
Router2(config-router)# no passive-interface FastEthernet0/1 # Para R3
```

Router3 Configuration:

```
Router3(config)# router ospf 1
Router3(config-router)# router-id 3.3.3.3
Router3(config-router)# network 10.23.10.0 0.0.0.3 area 0   # Link para R2
Router3(config-router)# network 10.34.10.0 0.0.0.3 area 0   # Link para R4
Router3(config-router)# network 10.20.1.4 0.0.0.3 area 0     # Company B
Router3(config-router)# passive-interface default
Router3(config-router)# no passive-interface FastEthernet0/0 # Para R2
Router3(config-router)# no passive-interface FastEthernet0/1 # Para R4
```

Router4 Configuration:

```
Router4(config)# router ospf 1
Router4(config-router)# router-id 4.4.4.4
Router4(config-router)# network 10.14.10.0 0.0.0.3 area 0   # Link para R1
Router4(config-router)# network 10.34.10.0 0.0.0.3 area 0   # Link para R3
Router4(config-router)# network 8.8.8.8 0.0.0.0 area 0       # Loopback0 único
Router4(config-router)# passive-interface default
Router4(config-router)# no passive-interface FastEthernet0/0 # Para R1
Router4(config-router)# no passive-interface FastEthernet0/1 # Para R3
```

OSPF Neighbor Relationships

```
# Router1 -----
Router1# show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:36   10.12.10.2   FastEthernet0/0
4.4.4.4          1     FULL/BDR        00:00:38   10.14.10.2   FastEthernet0/0

# Router2 -----
Router2# show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
1.1.1.1          1     FULL/BDR        00:00:32   10.12.10.1   FastEthernet0/0
3.3.3.3          1     FULL/BDR        00:00:33   10.23.10.2   FastEthernet0/1

# Router3 -----
Router3# show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
4.4.4.4          1     FULL/DR         00:00:33   10.34.10.2   FastEthernet0/0
2.2.2.2          1     FULL/DR         00:00:33   10.23.10.1   FastEthernet0/1

# Router4 -----
Router4# show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
1.1.1.1          1     FULL/BDR        00:00:39   10.14.10.1   FastEthernet0/0
3.3.3.3          1     FULL/BDR        00:00:31   10.34.10.1   FastEthernet0/1
```

OSPF Routes in Routing Table

```
Router1 -----
Gateway of last resort is 10.12.10.2 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 10.12.10.2, 00:12:19, FastEthernet0/0
         [110/2] via 10.14.10.2, 00:12:19, FastEthernet0/1
    10.0.0.0/30 is subnetted, 6 subnets
C       10.12.10.0 is directly connected, FastEthernet0/0
C       10.14.10.0 is directly connected, FastEthernet0/1
C       10.20.1.0 is directly connected, FastEthernet1/0
O       10.20.1.4 [110/3] via 10.12.10.2, 00:01:39, FastEthernet0/0
         [110/3] via 10.14.10.2, 00:01:39, FastEthernet0/1
O       10.23.10.0 [110/2] via 10.12.10.2, 00:01:59, FastEthernet0/0
O       10.34.10.0 [110/2] via 10.14.10.2, 00:01:39, FastEthernet0/1
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.20.11.0/24 [1/0] via 10.20.1.2
S       172.20.12.0/23 [1/0] via 10.20.1.2
    192.168.1.0/27 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 10.14.10.2
         [1/0] via 10.34.10.2
    192.168.2.0/27 is subnetted, 1 subnets
S       192.168.2.0 [1/0] via 10.14.10.2
         [1/0] via 10.34.10.2
S*    0.0.0.0/0 [1/0] via 10.12.10.2

Router2 -----
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
    10.0.0.0/30 is subnetted, 6 subnets
C       10.12.10.0 is directly connected, FastEthernet0/0
O       10.14.10.0 [110/2] via 10.12.10.1, 00:17:43, FastEthernet0/0
S       10.20.1.0 [1/0] via 10.12.10.1
S       10.20.1.4 [1/0] via 10.23.10.2
C       10.23.10.0 is directly connected, FastEthernet0/1
O       10.34.10.0 [110/2] via 10.23.10.2, 00:07:08, FastEthernet0/1
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S       172.20.11.0/24 [1/0] via 10.12.10.1
S       172.20.12.0/23 [1/0] via 10.12.10.1
    192.168.1.0/27 is subnetted, 1 subnets
S       192.168.1.0 [1/0] via 10.23.10.2
    192.168.2.0/27 is subnetted, 1 subnets
S       192.168.2.0 [1/0] via 10.23.10.2
S*    0.0.0.0/0 is directly connected, Loopback0

Router3 -----
Gateway of last resort is 10.23.10.1 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 10.23.10.1, 00:07:39, FastEthernet0/1
         [110/2] via 10.34.10.2, 00:07:39, FastEthernet0/0
    10.0.0.0/30 is subnetted, 6 subnets
O       10.12.10.0 [110/2] via 10.23.10.1, 00:07:59, FastEthernet0/1
O       10.14.10.0 [110/2] via 10.34.10.2, 00:07:39, FastEthernet0/0
S       10.20.1.0 [1/0] via 10.14.10.1
C       10.20.1.4 is directly connected, FastEthernet1/0
C       10.23.10.0 is directly connected, FastEthernet0/1
C       10.34.10.0 is directly connected, FastEthernet0/0
```

```

172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S    172.20.11.0/24 [1/0] via 10.34.10.2
S    172.20.12.0/23 [1/0] via 10.34.10.2
    192.168.1.0/27 is subnetted, 1 subnets
S    192.168.1.0 [1/0] via 10.20.1.6
    192.168.2.0/27 is subnetted, 1 subnets
S    192.168.2.0 [1/0] via 10.20.1.6
S*   0.0.0.0/0 [1/0] via 10.23.10.1

Router4 -----
Gateway of last resort is 10.14.10.1 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
C    8.8.8.8 is directly connected, Loopback0
    10.0.0.0/30 is subnetted, 6 subnets
O    10.12.10.0 [110/2] via 10.14.10.1, 00:18:43, FastEthernet0/0
C    10.14.10.0 is directly connected, FastEthernet0/0
S    10.20.1.0 [1/0] via 10.14.10.1
S    10.20.1.4 [1/0] via 10.34.10.1
O    10.23.10.0 [110/2] via 10.34.10.1, 00:08:03, FastEthernet0/1
C    10.34.10.0 is directly connected, FastEthernet0/1
    172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
S    172.20.11.0/24 [1/0] via 10.14.10.1
S    172.20.12.0/23 [1/0] via 10.14.10.1
    192.168.1.0/27 is subnetted, 1 subnets
S    192.168.1.0 [1/0] via 10.34.10.1
    192.168.2.0/27 is subnetted, 1 subnets
S    192.168.2.0 [1/0] via 10.34.10.1
S*   0.0.0.0/0 [1/0] via 10.14.10.1

```

5.1.2.2 Internet Simulation - Configuration of R2 Loopback0 = 8.8.8.8/32

It was already configured in Router2

```

Router2# show run
interface Loopback0
 ip address 8.8.8.8 255.255.255.255

```

5.1.2.3 Global connectivity with redundancy

Company A to "Internet" (8.8.8.8)

```

# From Company A Router -----
RouterA# ping 8.8.8.8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/19 ms

RouterA# traceroute 8.8.8.8
Tracing the route to 8.8.8.8

  1  10.20.1.1      0 msec   0 msec   0 msec
  2  10.14.10.2     0 msec   0 msec   0 msec

```

Company B to "Internet" (8.8.8.8)

```

# From Company B Router -----
RouterB# ping 8.8.8.8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

RouterB# traceroute 8.8.8.8
Tracing the route to 8.8.8.8

  1  10.20.1.5      0 msec   0 msec   0 msec
  2  10.34.10.2     0 msec   0 msec   0 msec

```

Inter-Company Communication

```

# From Company A to Company B internal network -----
RouterA# ping 192.168.1.1
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

RouterA# traceroute 192.168.1.1
Tracing the route to 192.168.1.1

  1  10.20.1.1      0 msec   0 msec   0 msec
  2  10.14.10.2     0 msec   0 msec   0 msec
  3  10.34.10.1     0 msec   1 msec   0 msec
  4  10.20.1.6      0 msec   0 msec   0 msec
  5  192.168.1.1    0 msec   0 msec  20 msec

# From Company B to Company A internal network -----
RouterB# ping 172.20.13.6
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/13 ms

```



```
RouterB# traceroute 172.20.13.6
Type escape sequence to abort.
Tracing the route to 172.20.13.6

 1  10.20.1.5      0 msec    0 msec    0 msec
 2  10.34.10.2     0 msec    0 msec    0 msec
 3  10.12.10.1     0 msec    0 msec    0 msec
 4  10.20.1.2      0 msec    0 msec    0 msec
 5  172.20.13.6    0 msec    0 msec    0 msec
```

Redundancy with Link Failure

Simulation a failure in the primary path between Router2 and Router1:

```
# On Router2 - shutdown interface to Router1
Router2(config)# interface Fa0/0
Router2(config-if)# shutdown
```

Verification of alternative paths:

```
# New route on Router1 after convergence -----
Router1#sh ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
4.4.4.4          1    2WAY/DROTHER    00:00:30    10.14.10.2     FastEthernet0/1

Router1#ping 8.8.8.8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router1#traceroute 8.8.8.8
Tracing the route to 8.8.8.8

 1  10.14.10.2     0 msec    0 msec    0 msec

# Test from Company B -----
RouterB# ping 8.8.8.8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/15 ms

# Test from Company A -----
RouterA# ping 8.8.8.8
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

After the test we restored the Router2 - Fa0/0 with `no shutdown` and all was restored as initially configured

5.1.3 Public addressing test

5.1.3.1 Change of configs so each company and the operator use public IPs

Public IP Address Allocation Plan

For simplicity, we decided to use the following **IP Addresses**:

- **200.X.10.Y** - ISP Core
- **200.20.X.Y** - Company A
- **200.168.X.Y** - Company B

Address Assignment Plan:

ISP Core Network:

- **Router1-Router2 Link (fa0/0):** 200.12.10.1/30
- **Router1-Router4 Link (fa0/1):** 200.14.10.1/30
- **Router1-RouterA Link:** 200.20.1.1/30
- **Router2 Loopback:** 8.8.8.8/32 (Internet simulation)
- **Router2-Router1 Link:** 200.12.10.2/30
- **Router2-Router3 Link:** 200.23.10.1/30
- **Router3-Router2 Link:** 200.23.10.2/30
- **Router3-Router4 Link:** 200.34.10.1/30
- **Router3-RouterB Link:** 200.20.1.5/30
- **Router4-Router1 Link:** 200.14.10.2/30
- **Router4-Router3 Link:** 200.34.10.2/30

Company A Internal Networks:

- **VLAN 11 (Accounting):** 200.20.11.0/24
- **VLAN 12 (Secretariat):** 200.20.12.0/24
- **VLAN 13 (Computer_science):** 200.20.13.0/25

Company B Internal Networks:

- **VLAN 2 (Servers):** 200.168.1.30/27
- **VLAN 3 (Engineering):** 200.168.2.30/27

ISP Core Reconfiguration

```

Router1 Reconfiguration -----
Router1(config)# interface FastEthernet0/0
Router1(config-if)# no ip address 10.12.10.1 255.255.255.252
Router1(config-if)# ip address 200.12.10.1 255.255.255.252

Router1(config)# interface FastEthernet0/1
Router1(config-if)# no ip address 10.14.10.1 255.255.255.252
Router1(config-if)# ip address 200.14.10.1 255.255.255.252

Router1(config)# interface FastEthernet1/0
Router1(config-if)# no ip address 10.20.1.1 255.255.255.252
Router1(config-if)# ip address 200.20.1.1 255.255.255.252

Router1(config)# ip route 200.20.11.0 255.255.255.0 200.20.1.2
Router1(config)# ip route 200.20.12.0 255.255.254.0 200.20.1.2

Router1(config)# ip route 200.168.1.0 255.255.255.224 200.14.10.2
Router1(config)# ip route 200.168.1.0 255.255.255.224 200.14.10.2

Router1(config)# ip route 0.0.0.0 0.0.0.0 200.12.10.2

# OSPF reconfiguration
Router1(config)# no router ospf 1
Router1(config)# router ospf 1
Router1(config-router)# router-id 1.1.1.1
Router1(config-router)# network 200.12.10.0 0.0.0.3 area 0      # Link para R2
Router1(config-router)# network 200.14.10.0 0.0.0.3 area 0      # Link para R4
Router1(config-router)# network 200.20.1.0 0.0.0.3 area 0      # Company A
Router1(config-router)# passive-interface default
Router1(config-router)# no passive-interface FastEthernet0/0    # Para R2
Router1(config-router)# no passive-interface FastEthernet0/1    # Para R4

Router2 Reconfiguration -----
Router2(config)# interface FastEthernet0/0
Router2(config-if)# no ip address 10.12.10.2 255.255.255.192
Router2(config-if)# ip address 200.12.10.2 255.255.255.252

Router2(config)# interface FastEthernet0/1
Router2(config-if)# no ip address 10.23.10.1 255.255.255.192
Router2(config-if)# ip address 200.23.10.1 255.255.255.252

Router2(config)# ip route 200.20.11.0 255.255.255.0 200.12.10.1
Router2(config)# ip route 200.20.12.0 255.255.254.0 200.12.10.1

Router2(config)# ip route 200.168.1.0 255.255.255.224 200.23.10.2
Router2(config)# ip route 200.168.1.0 255.255.255.224 200.23.10.2

# OSPF reconfiguration
Router2(config)# no router ospf 1
Router2(config)# router ospf 1
Router2(config-router)# router-id 2.2.2.2
Router2(config-router)# network 200.12.10.0 0.0.0.3 area 0      # Link para R1
Router2(config-router)# network 200.23.10.0 0.0.0.3 area 0      # Link para R3
Router2(config-router)# network 8.8.8.8 0.0.0.0 area 0          # Loopback0 (Internet)
Router2(config-router)# passive-interface default
Router2(config-router)# no passive-interface FastEthernet0/0    # Para R1
Router2(config-router)# no passive-interface FastEthernet0/1    # Para R3

Router3 Reconfiguration -----
Router3(config)# interface FastEthernet0/0
Router3(config-if)# no ip address 10.34.10.1 255.255.255.192
Router3(config-if)# ip address 200.34.10.1 255.255.255.192

Router3(config)# interface FastEthernet0/1
Router3(config-if)# no ip address 10.23.10.2 255.255.255.252
Router3(config-if)# ip address 200.23.10.2 255.255.255.252

Router3(config)# interface FastEthernet1/0
Router3(config-if)# no ip address 10.20.1.5 255.255.255.252
Router3(config-if)# ip address 200.20.1.5 255.255.255.252

Router3(config)# ip route 200.20.11.0 255.255.255.0 200.34.10.2
Router3(config)# ip route 200.20.12.0 255.255.254.0 200.34.10.2

Router3(config)# ip route 200.168.1.0 255.255.255.224 200.20.1.6
Router3(config)# ip route 200.168.2.0 255.255.255.224 200.20.1.6

Router3(config)# ip route 0.0.0.0 0.0.0.0 200.23.10.1

# OSPF reconfiguration
Router3(config)# no router ospf 1
Router3(config)# router ospf 1

```

```

Router3(config-router)# router-id 3.3.3.3
Router3(config-router)# network 200.23.10.0 0.0.0.3 area 0    # Link para R2
Router3(config-router)# network 200.34.10.0 0.0.0.3 area 0    # Link para R4
Router3(config-router)# network 200.20.1.4 0.0.0.3 area 0    # Company B
Router3(config-router)# passive-interface default
Router3(config-router)# no passive-interface FastEthernet0/0  # Para R2
Router3(config-router)# no passive-interface FastEthernet0/1  # Para R4

Router4 Reconfiguration -----
Router4(config)# interface FastEthernet0/0
Router4(config-if)# no ip address 10.14.10.2 255.255.255.252
Router4(config-if)# ip address 200.14.10.2 255.255.255.252

Router4(config)# interface FastEthernet0/1
Router4(config-if)# no ip address 10.34.10.2 255.255.255.252
Router4(config-if)# ip address 200.34.10.2 255.255.255.252

Router4(config)# ip route 0.0.0.0 0.0.0.0 200.14.10.1

Router4(config)# ip route 200.20.11.0 255.255.255.0 200.14.10.1
Router4(config)# ip route 200.20.12.0 255.255.254.0 200.14.10.1

Router4(config)# ip route 200.168.1.0 255.255.255.224 200.34.10.1
Router4(config)# ip route 200.168.2.0 255.255.255.224 200.34.10.1

# OSPF reconfiguration
Router4(config)# no router ospf 1
Router4(config)# router ospf 1
Router4(config-router)# router-id 4.4.4.4
Router4(config-router)# network 200.14.10.0 0.0.0.3 area 0    # Link para R1
Router4(config-router)# network 200.34.10.0 0.0.0.3 area 0    # Link para R3
Router4(config-router)# network 8.8.8.8 0.0.0.0 area 0        # Loopback0
Router4(config-router)# passive-interface default
Router4(config-router)# no passive-interface FastEthernet0/0  # Para R1
Router4(config-router)# no passive-interface FastEthernet0/1  # Para R3

```

Company A Reconfiguration (RouterA)

```

# WAN Interface
RouterA(config)# interface FastEthernet0/0
RouterA(config-if)# no ip address 10.20.1.2 255.255.255.252
RouterA(config-if)# ip address 200.20.1.2 255.255.255.252

# Subinterfaces for VLANs - Router-on-a-Stick
RouterA(config)# interface FastEthernet0/1.11
RouterA(config-subif)# no ip address 172.20.11.254 255.255.255.0
RouterA(config-subif)# ip address 200.20.11.254 255.255.255.0

RouterA(config)# interface FastEthernet0/1.12
RouterA(config-subif)# no ip address 172.20.12.254 255.255.255.0
RouterA(config-subif)# ip address 200.20.12.254 255.255.255.0

RouterA(config)# interface FastEthernet0/1.13
RouterA(config-subif)# no ip address 172.20.13.126 255.255.255.128
RouterA(config-subif)# ip address 200.20.13.126 255.255.255.128

# Default route
RouterA(config)# ip route 0.0.0.0 0.0.0.0 200.20.1.1

```

Company B Reconfiguration (RouterB)

```

# WAN Interface
RouterB(config)# interface FastEthernet0/0
RouterB(config-if)# no ip address 10.20.1.6 255.255.255.252
RouterB(config-if)# ip address 200.20.1.6 255.255.255.252

# Subinterfaces for VLANs
RouterB(config)# interface FastEthernet0/1.2
RouterB(config-subif)# no ip address 192.168.1.30 255.255.255.224
RouterB(config-subif)# ip address 200.168.1.30 255.255.255.224

RouterB(config)# interface FastEthernet0/1.3
RouterB(config-subif)# no ip address 192.168.2.30 255.255.255.224
RouterB(config-subif)# ip address 200.168.2.30 255.255.255.224

# Default route
RouterB(config)# ip route 0.0.0.0 0.0.0.0 200.20.1.5

```

End Device IPs

Company A PCs:

```

# PC7 & PC9 (Accounting VLAN 11)
PC7> ip 200.20.11.7 255.255.255.0 200.20.11.254

```

```
PC9> ip 200.20.11.9 255.255.255.0 200.20.11.254

# PCs & PC8 (Secretariat VLAN 12)
PC5> ip 200.20.12.5 255.255.255.0 200.20.12.254
PC8> ip 200.20.12.8 255.255.255.0 200.20.12.254

# PC6 (Computer_science VLAN 13)
PC6> ip 200.20.13.6 255.255.255.128 200.20.13.226
```

Company B PCs:

```
# Server (VLAN 2)
Server> ip 200.168.1.1 255.255.255.224 200.168.1.30

# Engineering PCs (VLAN 3)
PC1> ip 200.168.2.1 255.255.255.224 200.168.2.30
PC2> ip ip 200.168.2.2 255.255.255.224 200.168.2.30
```

Public Addressing Connectivity Tests

Basic ISP Core Connectivity

```
# From Router1
Router1# ping 200.12.10.2      # Router2
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1# ping 200.20.1.2      # RouterA
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router1# ping 200.20.1.6      # RouterB via OSPF
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1# ping 200.34.10.1     # Router3 via OSPF
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1# ping 200.14.10.2     # Router4
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1# ping 8.8.8.8         # Internet simulation
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

Inter-Company Communication

```
# Company A to Company B
RouterA# ping 200.20.1.6
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/18 ms

RouterA# traceroute 200.20.1.6
Tracing the route to 200.20.1.6

 0  200.20.1.1          0 msec    0 msec    0 msec
 1  200.12.10.2         0 msec    0 msec    0 msec
 2  200.23.10.2         0 msec    0 msec    0 msec
 3  200.20.1.6          0 msec    0 msec    0 msec
```

End-to-End from PCs

```
# From Company A PC6 to Internet
PC6> ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=30ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 30ms, Average = 7ms

# From Company A to Company B Server
PC6> ping 200.168.1.1
Pinging 200.168.1.1 with 32 bytes of data:

Reply from 200.168.1.1: bytes=32 time=1ms TTL=123
Reply from 200.168.1.1: bytes=32 time<1ms TTL=123
Reply from 200.168.1.1: bytes=32 time<1ms TTL=123
Reply from 200.168.1.1: bytes=32 time=1ms TTL=123

Ping statistics for 200.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5.2 Testing and Validation

5.2.1 show ip route, show ip ospf neighbor, show ip ospf interface brief

Router1

```
Router1>sh ip route

Gateway of last resort is 200.12.10.2 to network 0.0.0.0

      8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 200.12.10.2, 00:13:05, FastEthernet0/0
          [110/2] via 200.14.10.2, 00:13:05, FastEthernet0/1
      200.12.10.0/30 is subnetted, 1 subnets
C       200.12.10.0 is directly connected, FastEthernet0/0
      200.14.10.0/30 is subnetted, 1 subnets
C       200.14.10.0 is directly connected, FastEthernet0/1
      200.20.1.0/30 is subnetted, 2 subnets
C       200.20.1.0 is directly connected, FastEthernet1/0
O       200.20.1.4 [110/3] via 200.12.10.2, 00:18:28, FastEthernet0/0
S       200.20.11.0/24 [1/0] via 200.20.1.2
S       200.20.12.0/23 [1/0] via 200.20.1.2
      200.23.10.0/30 is subnetted, 1 subnets
O       200.23.10.0 [110/2] via 200.12.10.2, 00:18:38, FastEthernet0/0
      200.34.10.0/30 is subnetted, 1 subnets
O       200.34.10.0 [110/2] via 200.14.10.2, 00:13:05, FastEthernet0/1
      200.168.1.0/27 is subnetted, 1 subnets
S       200.168.1.0 [1/0] via 200.14.10.2
S*    0.0.0.0/0 [1/0] via 200.12.10.2

Router1>show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    FULL/BDR       00:00:34   200.12.10.2   FastEthernet0/0
4.4.4.4          1    FULL/BDR       00:00:34   200.14.10.2   FastEthernet0/1

Router1>show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    FULL/BDR       00:00:34   200.12.10.2   FastEthernet0/0
4.4.4.4          1    FULL/BDR       00:00:34   200.14.10.2   FastEthernet0/1
Router1>show ip ospf interface brief
Interface      PID   Area              IP Address/Mask      Cost   State   Nbrs F/C
Fa0/0          1     0                 200.12.10.1/255.255.252 1      DR     0/0
Fa0/1          1     0                 200.14.10.1/255.255.252 1      DR     0/0
Fa1/0          1     0                 200.20.1.1/255.255.252 1      WAIT   0/0
```

Router2

```
Router2>sh ip route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

      8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
      200.12.10.0/30 is subnetted, 1 subnets
C       200.12.10.0 is directly connected, FastEthernet0/0
      200.14.10.0/30 is subnetted, 1 subnets
O       200.14.10.0 [110/2] via 200.12.10.1, 00:18:17, FastEthernet0/0
      200.20.1.0/30 is subnetted, 2 subnets
O       200.20.1.0 [110/2] via 200.12.10.1, 00:31:27, FastEthernet0/0
O       200.20.1.4 [110/2] via 200.23.10.2, 00:23:40, FastEthernet0/1
S       200.20.11.0/24 [1/0] via 200.12.10.1
S       200.20.12.0/23 [1/0] via 200.12.10.1
      200.23.10.0/30 is subnetted, 1 subnets
C       200.23.10.0 is directly connected, FastEthernet0/1
      200.34.10.0/26 is subnetted, 1 subnets
O       200.34.10.0 [110/2] via 200.23.10.2, 00:23:50, FastEthernet0/1
      200.168.1.0/27 is subnetted, 1 subnets
S       200.168.1.0 [1/0] via 200.23.10.2
S*    0.0.0.0/0 is directly connected, Loopback0

Router2>sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          1    FULL/DR        00:00:39   200.12.10.1   FastEthernet0/0
3.3.3.3          1    FULL/BDR       00:00:34   200.23.10.2   FastEthernet0/1

Router2>sh ip ospf interface brief
Interface      PID   Area              IP Address/Mask      Cost   State   Nbrs F/C
Fa0/0          1     0                 200.12.10.2/255.255.252 1      BDR     0/0
Fa0/1          1     0                 200.23.10.1/255.255.252 1      DR     0/0
Lo0            1     0                 8.8.8.8/255.255.252 1      WAIT   0/0
```

Router3

```

Router3>sh ip route
Gateway of last resort is 200.23.10.1 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
O       8.8.8.8 [110/2] via 200.23.10.1, 00:24:42, FastEthernet0/1
    200.12.10.0/30 is subnetted, 1 subnets
O       200.12.10.0 [110/2] via 200.23.10.1, 00:24:42, FastEthernet0/1
    200.14.10.0/30 is subnetted, 1 subnets
O       200.14.10.0 [110/3] via 200.23.10.1, 00:19:51, FastEthernet0/1
    200.20.1.0/30 is subnetted, 2 subnets
O       200.20.1.0 [110/3] via 200.23.10.1, 00:24:42, FastEthernet0/1
C       200.20.1.4 is directly connected, FastEthernet1/0
S       200.20.11.0/24 [1/0] via 200.34.10.2
S       200.20.12.0/23 [1/0] via 200.34.10.2
    200.23.10.0/30 is subnetted, 1 subnets
C       200.23.10.0 is directly connected, FastEthernet0/1
    200.34.10.0/26 is subnetted, 1 subnets
C       200.34.10.0 is directly connected, FastEthernet0/0
    200.168.1.0/27 is subnetted, 1 subnets
S       200.168.1.0 [1/0] via 200.20.1.6
    200.168.2.0/27 is subnetted, 1 subnets
S       200.168.2.0 [1/0] via 200.20.1.6
S*    0.0.0.0/0 [1/0] via 200.23.10.1

Router3>sh ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    FULL/DR         00:00:30    200.23.10.1    FastEthernet0/1

Router3>sh ip ospf interface brief
Interface      PID   Area              IP Address/Mask      Cost   State   Nbrs F/C
Fa0/1          1     0                 200.23.10.2/255.255.255.252  1      BDR    0/0
Fa0/0          1     0                 200.34.10.1/255.255.255.192  1      DR     0/0
Fa1/0          1     0                 200.20.1.5/255.255.255.252  1      WAIT   0/0

```

Router4

```

Router4>sh ip route
Gateway of last resort is 200.14.10.1 to network 0.0.0.0

    8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
    200.12.10.0/30 is subnetted, 1 subnets
O       200.12.10.0 [110/2] via 200.14.10.1, 00:21:15, FastEthernet0/0
    200.14.10.0/30 is subnetted, 1 subnets
C       200.14.10.0 is directly connected, FastEthernet0/0
    200.20.1.0/30 is subnetted, 2 subnets
O       200.20.1.0 [110/2] via 200.14.10.1, 00:21:15, FastEthernet0/0
O       200.20.1.4 [110/4] via 200.14.10.1, 00:21:15, FastEthernet0/0
S       200.20.11.0/24 [1/0] via 200.14.10.1
S       200.20.12.0/23 [1/0] via 200.14.10.1
    200.23.10.0/30 is subnetted, 1 subnets
O       200.23.10.0 [110/3] via 200.14.10.1, 00:21:15, FastEthernet0/0
    200.34.10.0/30 is subnetted, 1 subnets
C       200.34.10.0 is directly connected, FastEthernet0/1
    200.168.1.0/27 is subnetted, 1 subnets
S       200.168.1.0 [1/0] via 200.34.10.1
    200.168.2.0/27 is subnetted, 1 subnets
S       200.168.2.0 [1/0] via 200.34.10.1
S*    0.0.0.0/0 [1/0] via 200.14.10.1

Router4>sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          1    FULL/DR         00:00:37    200.14.10.1    FastEthernet0/0

Router4>sh ip ospf interface brief
Interface      PID   Area              IP Address/Mask      Cost   State   Nbrs F/C
Fa0/0          1     0                 200.14.10.2/255.255.255.252  1      BDR    0/0
Fa0/1          1     0                 200.34.10.2/255.255.255.252  1      DR     0/0
Lo0            1     0                 8.8.8.8/255.255.255.255    1      WAIT   0/0

```

5.2.2 ping/traceroute from company PCs to ISP and to 8.8.8.8

```

# From Company A
RouterA>ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/14 ms

RouterA>traceroute 8.8.8.8
Type escape sequence to abort.
Tracing the route to 8.8.8.8

 0  1  200.20.1.1      12 msec  0 msec  0 msec

```

```

2    200.12.10.2    0 msec    0 msec    0 msec

# From Company B
RouterB>ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

RouterB>traceroute 8.8.8.8
Type escape sequence to abort.
Tracing the route to 8.8.8.8

 0  100.1.1.1         0 msec    0 msec    0 msec
 1  200.20.1.5        0 msec    0 msec    0 msec
 2  200.23.10.1       0 msec    0 msec    0 msec

```

5.3 Practical Questions

1. Can corporate PCs ping the ISP router before OSPF? Justify.

Yes, with exception of PC7 and PC9 from Company A, because the sub-interface for VLAN 11 in the Company Router is shutdown. All other PCs can ping ISP routers because in the Company Routers there is a default route pointing to them and then in the ISP routers there are routes pointing back to the Companies

2. What is the next-hop of your default routes and why?

Router	next-hop	Justification
RouterA	10.20.1.1	Router1 Fa1/0 - ISP closest router
RouterB	10.20.1.5	Router3 Fa1/0 - ISP closest router
Router1	10.12.10.2	Router2 Fa0/0 - Connects to Loopback0 internally
Router2	directly connected	Loopback0 (Internet simulation)
Router3	10.23.10.1	Router2 Fa0/1 - Connects to Loopback0 internally
Router4	10.14.10.1	Router1 F0/1 - Has internal route to default too. Could have been 10.34.10.1, the result will be the same

table 24 - next-hop per Router

3. Describe the purpose of static routes on R1–R4 and what changes once OSPF runs.

Static routes on R1-R4 provide **manual path instruction** to create complete connectivity across the ISP fabric when dynamic routing is unavailable.

The transition from static to OSPF routing transforms the network from a **manually configured mesh** to an **intelligent, self-healing fabric** while maintaining appropriate boundaries between ISP internal routing and customer connectivity.

4. After enabling OSPF, explain the path selection you observe towards 8.8.8.8.

OSPF uses **cost** as its metric, which is calculated as:

Cost = Reference Bandwidth (100 Mbps) / Interface Bandwidth

For FastEthernet (100 Mbps): $100 / 100 = \text{Cost } 1$

For Loopback interfaces: $\text{Cost} = 1$ (very low)

Path Cost Comparison Table:

Source Router	Possible Paths	Total Cost	Chosen Path
Router1	Direct to R2	2	R1 → R2
Router3	R3 → R4 → R2	3	R3 → R4 → R2
	R3 → R1 → R2	N/A	Not possible (no direct link)
Router4	Direct to R2	2	R4 → R2

table 25 - OSPF path costs

The OSPF path selection toward **8.8.8.8** demonstrates:

1. **Optimality:** Always chooses the lowest-cost path
2. **Direct Connections Preferred:** Router1 and Router4 use direct links to Router2
3. **Transit Routing:** Router3 must use Router4 as transit to reach Router2
4. **Deterministic:** Same topology always produces the same paths
5. **Scalable:** Adding more routers would automatically calculate new optimal paths

This proves OSPF is successfully replacing manual static routing with intelligent, dynamic path selection based on the actual network topology and link costs.

Conclusion

This project successfully bridges theoretical networking concepts with real-world implementation, demonstrating that a methodical, phased approach to network design yields robust, enterprise-grade infrastructure. The seamless integration of Layer 2 segmentation, inter-VLAN routing, dynamic path selection, and multi-tenant ISP services validates the comprehensive understanding of modern networking principles.

The implemented solution not only meets all specified technical requirements but also establishes a foundation that exemplifies industry best practices in network design, security, and reliability. This project serves as a testament to the power of systematic network engineering in building infrastructures that are secure, scalable, and resilient—capable of supporting modern organizational needs while providing a platform for future technological evolution.

The knowledge and skills demonstrated through this implementation provide a solid foundation for designing, deploying, and managing complex network infrastructures in enterprise environments, ensuring optimal performance, security, and reliability for business-critical applications and services.